

Team: Snow Build

My CoCo Build Journey - Snowflake CoCo CLI Hackathon (GCC Edition)

Team Leader: Vipul Maurya | Problem #5: Supply Chain Ontology and Governed Conversational Analytics | Account: ZHSBCWU-TN70880 (AWS ap-south-1) | User: VIPULMAURYA

A step-by-step journal of how I used Snowflake CoCo to plan, build, run and test my solution.

Journey at a glance

#	Date	Phase	CoCo surface	Prompt	Result
1	28 Sep 2026	Setup / Planning	Snowsight (CoCo panel)	What databases do I have access to?	Success - listed 4 databases: SNOWFLAKE, SNOWFLAKE_LEARNING_DB, SNOWFLAKE_SAMPLE_DATA, USER\$VIPULMAURYA
2	28 Sep 2026	Setup / Planning	CoCo CLI v1.1.87 (PowerShell)	What databases do I have access to?	Success - sql-author skill + SQL_EXECUTE on zhsbcwu-tn70880; listed 4 databases. Model: Auto (Efficient)
3	28 Sep 2026	Development (synthetic data)	CoCo CLI	Create warehouse HACK_WH (XSMALL, auto-suspend 60s), database HACK_DEMO, schema	Success - HACK_WH, HACK_DEMO, RAW, CUSTOMERS (100 rows) created. CoCo auto-fixed 2 SQL compilation

				RAW, CUSTOMERS table with 100 synthetic rows; show top 10	errors (RANDOM/U NIFORM constant args) and retried.
4	28 Sep 2026	Testing / validation	Snowsight Catalog	Verify HACK_DEMO .RAW.CUSTOMERS created by CoCo CLI	Verified - table exists in Catalog with 100 rows and 10 columns; data profile shows distributions per column.
5	28 Sep 2026	Planning	CoCo CLI	Start new project session in sc_ontology folder	Fresh session, connection zhsbcwu-tn70880, model auto-efficient
6	28 Sep 2026	Planning	CoCo CLI (plan mode)	Planning prompt for Problem #5 Supply Chain Ontology	CoCo checked region, cross-region inference (ANY_REGION) and AI access; found missing AI permission
7	28 Sep 2026	Planning / Setup	Snowsight	GRANT USE AI FUNCTIONS ON ACCOUNT TO ROLE ACCOUNTADMIN	Permission gap found by CoCo fixed - statement executed successfully
8	28 Sep 2026	Planning	CoCo CLI	Answer CoCo clarifying questions	Added OTIF, Supplier Lead Time Var, Backorder Rate; Claude model for agent; native

					SQL data generation
9	28 Sep 2026	Planning (review)	CoCo CLI	Review proposed plan and send 12 corrections	CoCo updated design: cost-safe schedules, complete RAW/CURATED layers, fixed metric formulas
10	28 Sep 2026	Planning (output)	CoCo CLI	Save plan as PLAN.md and stop	PLAN.md (400 lines, 12 sections) saved; CoCo waiting for Phase 1 approval
11	28 Sep 2026	Development (setup)	CoCo CLI	Phase 1 prompt: database, schemas, RAW tables, generator, LEGACY views	SC_ONTOLONGY with RAW/LEGACY/CURATED/SEMANTIC/APP, 19 RAW tables incl. FX_RATES and UOM_CONVERSIONS; SQL saved to sql/01-04
12	28 Sep 2026	Development + Testing	CoCo CLI	Generate synthetic data and verify demo patterns	CoCo fixed 2 SQL errors, detected off-target anomalies and recalibrated; review found Planning view at 100% -> fixed to 83.51%
13	28 Sep 2026	Testing (validation)	CoCo CLI	Final Phase 1 verification summary	All targets met: SUP-104 OTIF 64% / +14 days, 4-day stockout, CARR-04 OTD 54.42%,

					legacy OTD 83.51 / 96.65 / 64.12%
14	28 Sep 2026	Execution (cost control)	Snowsight	Check spend after Phase 1	About USD 0.8 used this month of the USD 400 trial; warehouses auto- suspend after 60 s

Step 1 - CoCo connection test (Snowsight)

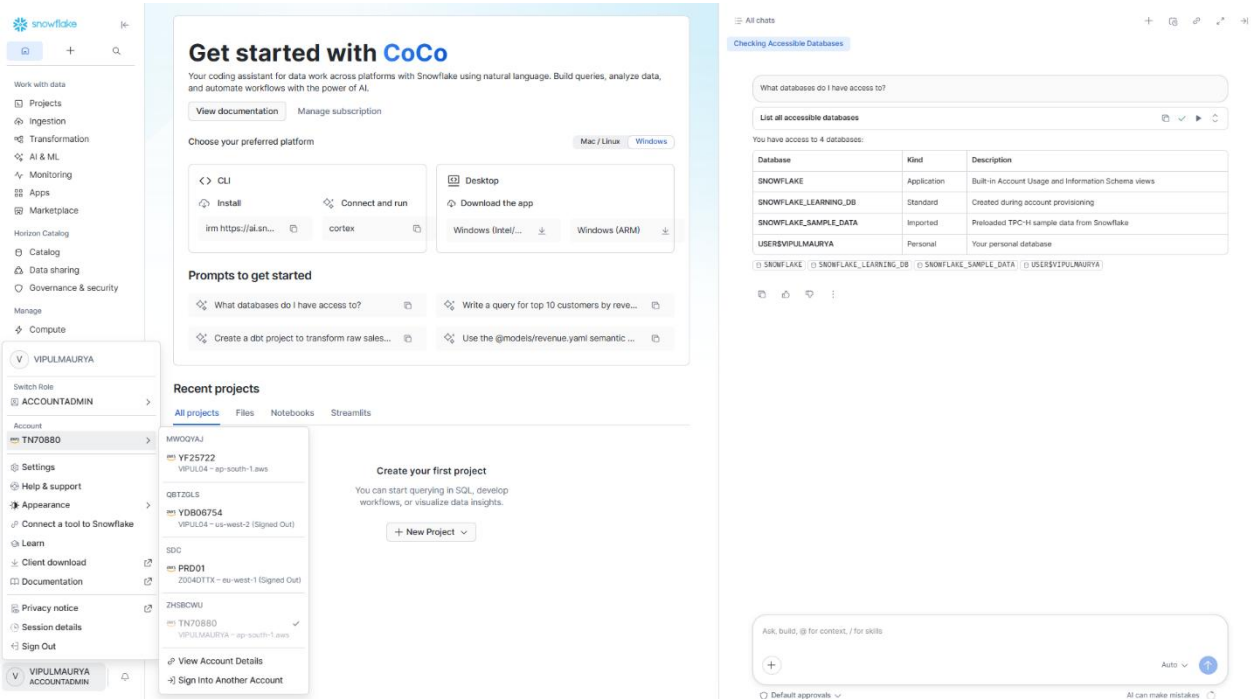
Date: 28 Sep 2026

Surface: CoCo in Snowsight, account ZHSBCWU-TN70880

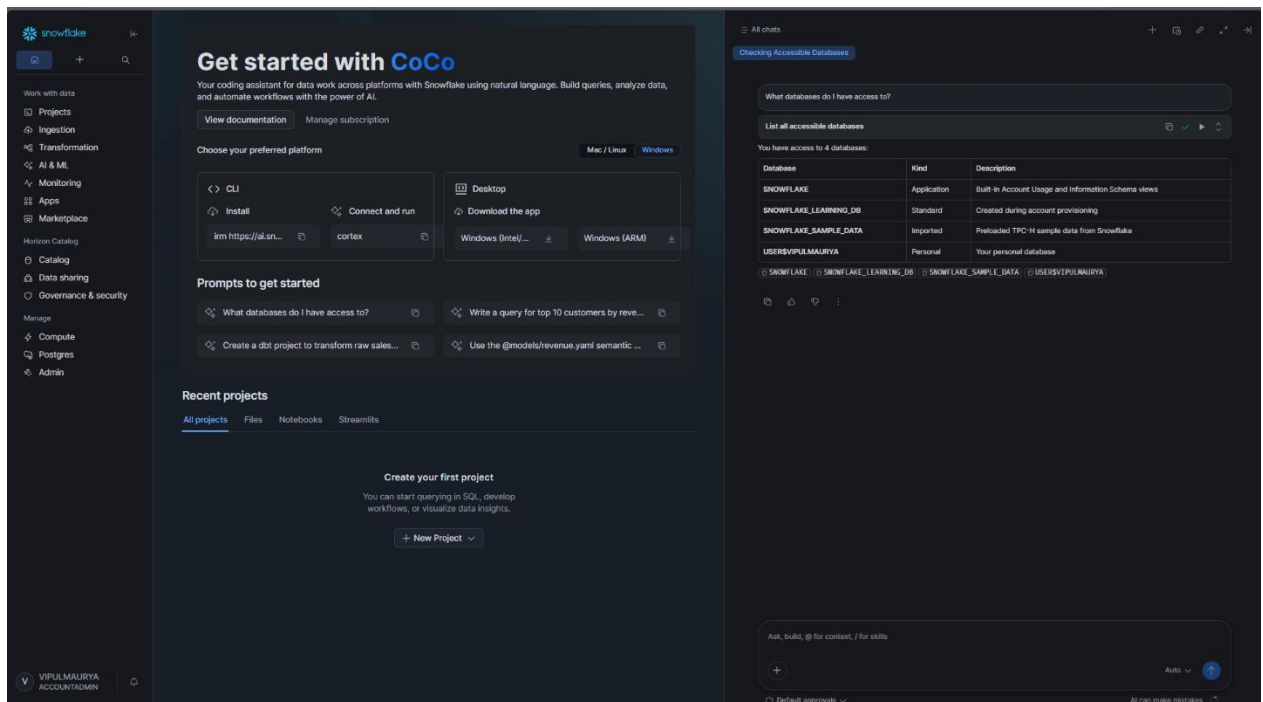
Prompt: What databases do I have access to?

Result: CoCo ran 'List all accessible databases' and returned 4 databases.

Note: the original standard trial (MWOQYAJ-YF25722) blocked CoCo ('Access denied for trial accounts'), so a dedicated CoCo trial account was created.



Screenshot: 01_CoCo_Snowsight_databases_account_TN70880.png



Screenshot: 02_CoCo_Snowsight_databases_dark.png

Step 2 - CoCo connection test (CLI)

Date: 28 Sep 2026

Surface: CoCo CLI v1.1.87 in PowerShell, started with: `cortex -c zhshbcwu-tn70880`

Agent + SQL connection: zhshbcwu-tn70880 | Model: Auto (Efficient)

Prompt: What databases do I have access to?

Result: CoCo used the `sql-author` skill, ran `SQL_EXECUTE 'Show all accessible databases'` and returned `SNOWFLAKE`, `SNOWFLAKE_LEARNING_DB`, `SNOWFLAKE_SAMPLE_DATA`, `USER$VIPULMAURYA`.

The screenshot shows the Snowflake CoCo CLI interface. At the top, a 'Session Summary' box displays session details. Below it, the 'Cortex Code v1.1.87' section shows a terminal window with a Snowflake logo and connection information. The main terminal area shows the command 'cortex --resume' and the output of the 'SQL_EXECUTE' command, which lists accessible databases. A table titled 'SQL Result (4 rows x 14 cols)' shows the following data:

#	created_on	name	is_default	is_current	origin	owner	comment
1	"2026-09-27 23:30:58.601 -0700"	SNOWFLAKE	N	N	SNOWFLAKE, ACCOUNT_USAGE	ACCOUNTADMIN	Created by Snowflake during account provi...
2	"2026-09-27 23:31:02.189 -0700"	SNOWFLAKE_LEARNING_DB	N	N		ACCOUNTADMIN	
3	"2026-09-27 23:31:02.197 -0700"	SNOWFLAKE_SAMPLE_DATA	N	N	SFSALESHARED.SFC_SAMPLES_AWSAPSOUTH1.SAMPLE_DATA	ACCOUNTADMIN	Preloaded TPC-H Data provided by Snowflake...

Below the table, the text 'You have access to the following databases:' is followed by a list of databases: SNOWFLAKE (System Application / Account Usage), SNOWFLAKE_LEARNING_DB (Standard Database), SNOWFLAKE_SAMPLE_DATA (Imported Sample Database), and USER\$VIPULMAURYA (Personal Database). The bottom of the terminal shows the 'confirm actions' and 'view: compact' options.

Screenshot: 03_CoCo_CLI_databases_account_TN70880.png

Step 3 - Warehouse, database and synthetic CUSTOMERS data (CLI)

Date: 28 Sep 2026 | Surface: CoCo CLI | Connection: zhsbcwu-tn70880 | Model: Auto (Efficient)

Prompt: Create a warehouse HACK_WH (XSMALL, auto-suspend 60 seconds), a database HACK_DEMO with schema RAW, and a CUSTOMERS table with 100 realistic synthetic rows. Then show me the top 10 rows.

Steps executed by CoCo (SQL_EXECUTE): 1) Create warehouse HACK_WH 2) Create database HACK_DEMO 3) Create schema RAW 4) Create CUSTOMERS table with 100 synthetic rows 5) Select top 10 rows.

Error handling: step 4 failed twice with SQL compilation errors (RANDOM needs a constant argument; UNIFORM needs constant bounds). CoCo diagnosed each error, rewrote the SQL and succeeded on the third attempt without manual intervention.

Result: HACK_DEMO.RAW.CUSTOMERS with 10 columns (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, CITY, COUNTRY, SIGNUP_DATE, LOYALTY_STATUS, TOTAL_SPENT).

Observation for testing phase: some City/Country pairs are inconsistent (e.g. Melbourne - United States, Munich - Australia); to be validated/fixed with CoCo.

```
Windows PowerShell
+ SNOWFLAKE_LEARNING_DB (Standard Database)
+ SNOWFLAKE_SAMPLE_DATA (Imported Sample Database)
+ USER$VIZULMAURYA (Personal Database)

Recap: We identified your accessible Snowflake databases. Let me know which database or tables you want to explore next. (You can turn recaps off in /settings)

* Create a warehouse HACK_WH (XSMALL, auto-suspend 60 seconds), a database HACK_DEMO with schema RAW, and a CUSTOMERS table with 100 realistic synthetic rows. Then show me the top 10 rows.

* I will create the warehouse HACK_WH, database HACK_DEMO, schema RAW, and populate CUSTOMERS with 100 synthetic rows.

✓ [SQL_EXECUTE] Create warehouse HACK_WH
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 1 cols)

# | status
--|-----
1 | Warehouse HACK_WH successfully created.

* I will create the database HACK_DEMO and schema RAW.

✓ [SQL_EXECUTE] Create database HACK_DEMO
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 1 cols)

# | status
--|-----
1 | Database HACK_DEMO successfully created.

* I will create the schema RAW inside HACK_DEMO.

✓ [SQL_EXECUTE] Create schema RAW in HACK_DEMO
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 1 cols)

# | status
--|-----
1 | Schema RAW successfully created.

* I will create the CUSTOMERS table and insert 100 realistic synthetic rows.

✗ [SQL_EXECUTE] Create CUSTOMERS table with 100 synthetic rows
Connection: zhsbcwu-tn70880
SQL compilation error:
  argument 1 to function RANDOM needs to be constant, found 'ROW_NUMBER() OVER (ORDER BY null ASC NULLS LAST)'
```

✗ I will generate the synthetic customer records using RANDOM() without dynamic seeds.

✗ [SQL_EXECUTE] Create CUSTOMERS table with 100 synthetic rows
Connection: zhsbcwu-tn70880
SQL compilation error:
argument 2 to function UNIFORM needs to be constant, found '(ARRAY_SIZE(FN.ARR)) - 1'

✗ I will create the table with constant array index bounds.

✓ [SQL_EXECUTE] Create CUSTOMERS table with 100 synthetic rows

Screenshot: 04_CoCo_CLI_create_HACK_WH_DB_schema_selfcorrect.png

```
Windows PowerShell

* I will create the table with constant array index bounds.

✓ [SQL_EXECUTE] Create CUSTOMERS table with 100 synthetic rows
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 1 cols)

# | status
--|-----
1 | Table CUSTOMERS successfully created.

* I will fetch the top 10 rows from HACK_DEMO.RAW.CUSTOMERS.

✓ [SQL_EXECUTE] Select top 10 customers from HACK_DEMO.RAW.CUSTOMERS
Connection: zhsbcwu-tn70880
SQL Result (10 rows x 10 cols) [ctrl+t to navigate]

# | CUSTOMER_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | CITY | COUNTRY | SIGNUP_DATE | LOYALTY_STATUS | TOTAL_SPENT
--|-----
1 | 1 | Jennifer | Clark | jennifer.clark1@gmail.com | +58-668-4459388 | Sydney | Australia | "2026-09-02" | Silver | 2349.00
2 | 2 | Mary | Thompson | mary.thompson2@icloud.com | +44-747-9718584 | Melbourne | United States | "2026-03-09" | Bronze | 1037.00
3 | 3 | David | Perez | david.perez3@icloud.com | +93-806-5947994 | Melbourne | United States | "2024-09-09" | Bronze | 1642.00

Rows 1-3/10

* The warehouse HACK_WH, database HACK_DEMO, schema RAW, and table CUSTOMERS (with 100 synthetic rows) have been created.

Top 10 Rows from HACK_DEMO.RAW.CUSTOMERS

CUSTOMER_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | CITY | COUNTRY | SIGNUP_DATE | LOYALTY_STATUS | TOTAL_SPENT
--|-----
1 | Jennifer | Clark | jennifer.clark1@gmail.com | +58-668-4459388 | Sydney | Australia | 2026-09-02 | Silver | $2,349.00
2 | Mary | Thompson | mary.thompson2@icloud.com | +44-747-9718584 | Melbourne | United States | 2026-03-09 | Bronze | $1,037.00
3 | David | Perez | david.perez3@icloud.com | +93-806-5947994 | Melbourne | United States | 2024-09-09 | Bronze | $1,642.00
4 | Patricia | Sanchez | patricia.sanchez4@yahoo.com | +61-434-9835483 | London | United States | 2026-08-04 | Bronze | $2,178.00
5 | Karen | Jones | karen.jones5@outlook.com | +96-178-4858693 | Philadelphia | United States | 2024-09-09 | Platinum | $2,662.00
6 | Michael | Smith | michael.smith6@yahoo.com | +94-178-2654518 | Munich | Australia | 2025-11-19 | Gold | $4,416.00
7 | Patricia | Robinson | patricia.robinson7@yahoo.com | +28-487-5829011 | Phoenix | Australia | 2025-10-18 | Silver | $642.00
8 | Anthony | Johnson | anthony.johnson8@yahoo.com | +57-748-7514318 | New York | United States | 2026-06-14 | Gold | $3,680.00
9 | Richard | Smith | richard.smith9@icloud.com | +44-362-9282347 | San Diego | Australia | 2025-10-28 | Platinum | $1,466.00
10 | Anthony | Lee | anthony.lee10@example.com | +39-818-1589266 | Vancouver | United Kingdom | 2026-02-17 | Platinum | $2,175.00

Type your message...

Understanding What Databases You Have Access To

** confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+e) @zhsbcwu-tn70880
```

Screenshot: 05_CoCo_CLI_CUSTOMERS_100_rows_top10.png

Step 4 - Cross-surface verification in Snowsight

Date: 28 Sep 2026 | Surface: Snowsight > Catalog > TN70880 > HACK_DEMO > RAW > CUSTOMERS > Preview

Check: objects built by CoCo CLI (Step 3) are visible in Snowsight. Result: CUSTOMERS table present with 100 rows; column profiles visible (e.g. LOYALTY_STATUS, COUNTRY distribution).

The screenshot displays the Snowflake Snowsight interface. On the left, the 'Catalog explorer' shows the hierarchy: TN70880 > HACK_DEMO > RAW > CUSTOMERS. The main area shows the 'CUSTOMERS' table with 100 rows and 14 columns. The columns are: CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, CITY, COUNTRY, BIRTH_DATE, LOYALTY_STATUS, and others. The table is displayed in a grid view. On the right, there is a sidebar with 'Checking Accessible Databases' and a list of accessible databases: SNOWFLAKE, SNOWFLAKE_LEARNING_DB, SNOWFLAKE_SAMPLE_DATA, and USERVPLUMAUITY. The bottom right corner has a chat window with the prompt 'Ask, build, @ for context, / for skills'.

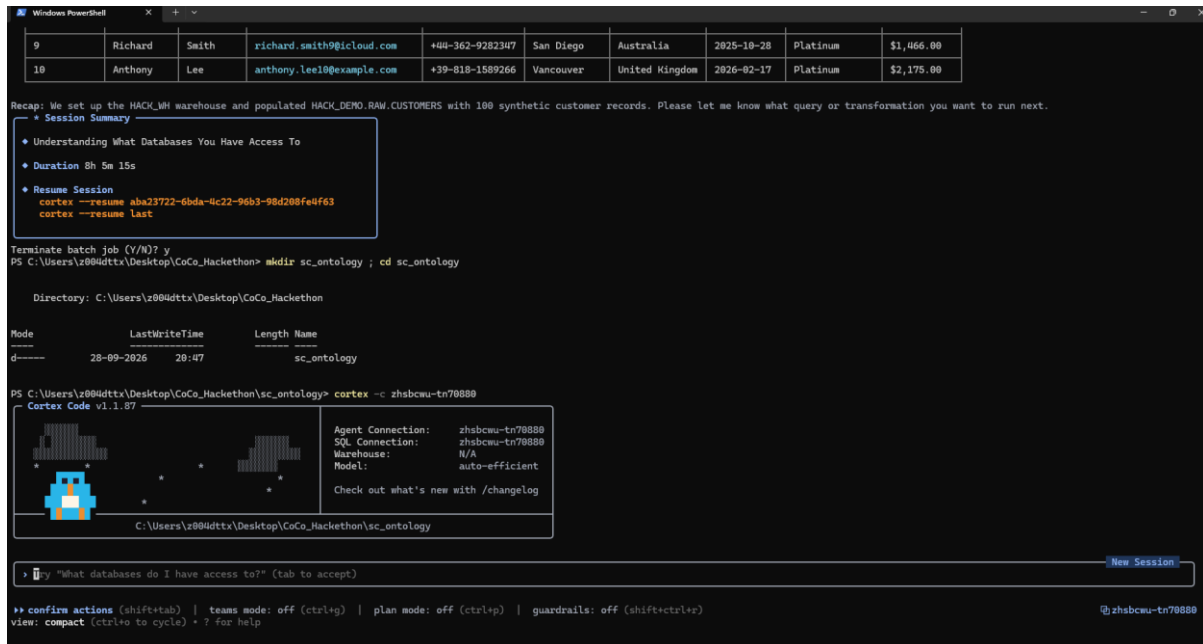
CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	CITY	COUNTRY	BIRTH_DATE	LOYALTY_STATUS
1	Jeffrey	Clark	jeffrey.clark1@gmail.com	+58-669-449308	Sydney	Australia	2026-09-02	Silver
2	Mary	Thompson	mary.thompson3@icloud.com	+44-747-971854	Melbourne	United States	2026-03-09	Bronze
3	David	Perez	david.perez3@icloud.com	+93-806-594794	Melbourne	United States	2024-09-09	Bronze
4	Patricia	Sanchez	patricia.sanchez4@yahoo.com	+61-434-803543	London	United States	2026-08-04	Bronze
5	Karen	Jones	karen.jones4@outlook.com	+96-178-485063	Philadelphia	United States	2024-09-09	Platinum
6	Michael	Smith	michael.smith@outlook.com	+94-178-2654518	Munich	Australia	2025-11-19	Gold
7	Patricia	Robinson	patricia.robinson7@yahoo.com	+20-487-5830011	Phoenix	Australia	2025-10-18	Silver
8	Anthony	Johnson	anthony.johnson8@yahoo.com	+57-748-7914318	New York	United States	2026-06-14	Gold
9	Ronald	Smith	ronald.smith9@icloud.com	+44-362-6922347	San Diego	Australia	2025-10-28	Platinum
10	Anthony	Lee	anthony.lee10@icloud.com	+59-818-1580568	Vancouver	United Kingdom	2026-03-17	Platinum
11	David	Johnson	david.johnson11@yahoo.com	+20-362-1144135	Manchester	United Kingdom	2026-04-04	Platinum
12	Nancy	Martin	nancy.martin12@example.com	+42-107-7985733	Toronto	United States	2025-09-22	Gold
13	Jessica	Martin	jessica.martin13@yahoo.com	+85-276-6842178	Toronto	France	2026-07-28	Platinum
14	Karen	Sanchez	karen.sanchez14@icloud.com	+21-991-2412224	Dallas	Canada	2026-06-17	Gold
15	Barbara	Clark	barbara.clark15@icloud.com	+33-117-2142937	Lyon	Germany	2026-05-13	Platinum
16	Karen	Robinson	karen.robinson16@yahoo.com	+88-615-9728921	Los Angeles	United Kingdom	2025-08-31	Gold
17	Sandra	Thompson	sandra.thompson17@gmail.com	+56-967-1345290	Phoenix	Canada	2024-01-15	Silver
18	Robert	Thompson	robert.thompson18@outlook.co	+68-902-6572299	Vancouver	United Kingdom	2025-11-18	Gold
19	Elizabeth	Perez	elizabeth.perez19@icloud.com	+16-568-1395513	Los Angeles	United Kingdom	2024-08-11	Bronze
20	Margaret	Harris	margaret.harris20@yahoo.com	+34-604-5447149	Melbourne	Germany	2024-04-18	Bronze
21	Thomas	Martin	thomas.martin21@yahoo.com	+93-214-7026712	Paris	United States	2024-06-30	Gold
22	Anthony	Thomas	anthony.thomas22@gmail.com	+2-985-3000532	New York	Germany	2025-03-07	Bronze
23	William	Ramirez	william.ramirez23@example.co	+15-765-6664745	Vancouver	France	2024-11-03	Bronze

Screenshot: 06_Snowsight_verify_CUSTOMERS_100_rows.png

Step 5 - New project session for the real build (CLI)

Date: 28 Sep 2026 | Surface: CoCo CLI v1.1.87 | Folder: CoCo_Hackathon\sc_ontology |
Connection: zhsbcwu-tn70880

Why: closed the practice session and started a clean CoCo session inside a dedicated project folder, so all generated files (plans, SQL, app code) are saved together and old chat history does not add cost.



```
Windows PowerShell
+ ~~~~~
9 | Richard | Smith | richard.smith@icloud.com | +04-362-9282347 | San Diego | Australia | 2025-10-28 | Platinum | $1,466.00
10 | Anthony | Lee | anthony.lee10@example.com | +39-010-1589266 | Vancouver | United Kingdom | 2026-02-17 | Platinum | $2,175.00

Recap: We set up the HACK_WH warehouse and populated HACK_DEMO.RAW.CUSTOMERS with 100 synthetic customer records. Please let me know what query or transformation you want to run next.

* Session Summary
  * Understanding What Databases You Have Access To
  * Duration 8h 5m 15s
  * Resume Session
    cortex --resume aha23722-6bda-4c22-96b3-98d208fe4f63
    cortex --resume Last

Terminate batch job (Y/N)? y
PS C:\Users\z004dtt\Desktop\CoCo_Hackathon> mkdir sc_ontology ; cd sc_ontology

Directory: C:\Users\z004dtt\Desktop\CoCo_Hackathon

Mode                LastWriteTime         Length Name
----                -
d-----          28-09-2026    20:47             sc_ontology

PS C:\Users\z004dtt\Desktop\CoCo_Hackathon\sc_ontology> cortex -c zhsbcwu-tn70880

Cortex Code v1.1.87

Agent Connection: zhsbcwu-tn70880
SQL Connection:  zhsbcwu-tn70880
Warehouse:      N/A
Model:          auto-efficient
Check out what's new with /changelog

C:\Users\z004dtt\Desktop\CoCo_Hackathon\sc_ontology

> | "What databases do I have access to?" (tab to accept)

New Session

>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+z)
view: compact (ctrl+o to cycle) + ? for help
```

Screenshot: 07_CoCo_CLI_new_project_session_sc_ontology.png

Step 6 - Planning prompt and environment checks by CoCo

Date: 28 Sep 2026 | Surface: CoCo CLI (plan mode) | Chosen problem: #5 Supply Chain Ontology and Governed Conversational Analytics

Prompt (summary): act as solution architect, **PLANNING ONLY**, no Snowflake objects yet; fictional manufacturer Apex Industrial Components (4 plants, 30 suppliers, 200 parts, 50 products, 100 customers, 12 months); ontology Supplier-Part-Plant-Inventory-Product-Sales Order-Shipment-Customer; canonical metrics OTD, Fill Rate, Days of Inventory, Landed Cost; deliver PLAN.md with 12 sections and check Cortex availability in AWS ap-south-1.

What CoCo did: before planning, it ran read-only checks on its own - account region (AWS_AP_SOUTH_1), role (ACCOUNTADMIN), cross-region inference parameter (already ANY_REGION) and available Cortex database roles. Each SQL step was approved manually.

Issue found: the LLM availability test failed with 'Unknown function COMPLETE' - the role had not been granted permission to use AI functions. CoCo reported the exact GRANT needed.

```

- Cortex Search over supplier contract / SLA documents
- Cortex Agent (Cortex Analyst on the semantic view + Cortex Search) with a custom tool (stored procedure) that raises a supplier escalation / action item into APP.ACTIONS
- Streamlit in Snowflake app: MPI dashboard, metric glossary, ontology explorer, persona chat (Planning / Procurement / Logistics), "Before vs After" consistency page, actions log
- Scheduled task: daily refresh + automated metric consistency test
- Guardrails: agent asks for clarification on ambiguous questions, shows metric definition with each answer, refuses questions outside the ontology, data quality checks

## Personas and proof of consistency
- Planner, Procurement Manager, Logistics Manager each ask the same metric question in their own words and must receive the identical number with the same definition.

## Deliverable of this step
Create a file PLAN.md in the current folder containing:
1. Problem framing and success criteria
2. Demo story: 5 questions (persona -> question -> answer -> action), including the "before vs after" moment
3. Ontology: entities, relationships, hierarchies (Mermaid ER diagram)
4. Table-by-table columns for RAW, LEGACY and CURATED layers (with keys)
5. Canonical metric glossary: name, business definition, exact SQL formula, grain, owner
6. Synthetic data plan: row counts, distributions, injected inconsistencies and patterns
7. Pipeline design (which objects are dynamic tables / streams / tasks and why)
8. Semantic view design: tables, dimensions, facts, metrics, synonyms, 10 verified queries
9. Agent design: tools, instructions, guardrails, example questions
10. Streamlit app pages and user actions
11. Test and validation plan (persona consistency test, accuracy checks, edge cases, data quality)
12. Build phases in order with estimated credit cost per phase
Also check whether Cortex Analyst, Cortex Search, Cortex Agents and the needed LLM models are available in my region (AWS ap-south-1) and tell me if any account setting (e.g., cross-region inference) is required.

* I will check the Snowflake account region, Cortex Analyst/Search/Agent support, and cross-region inference settings.

/ SQL_EXECUTE Check Snowflake account region, edition, and parameter se...
Connection: zhbcwu-tn70880
SQL Result (1 rows x 5 cols)

+-----+-----+-----+-----+-----+
| # | CURRENT_REGION() | CURRENT_ACCOUNT() | CURRENT_VERSION() | CURRENT_ROLE() | CURRENT_WAREHOUSE() |
+-----+-----+-----+-----+-----+
| 1 | AWS_AP_SOUTH_1 | SK96811 | 10.35.101 | ACCOUNTADMIN | COMPUTE_WH |
+-----+-----+-----+-----+-----+

Execute SQL
SHOW PARAMETERS LIKE 'CORTEX_ENABLED_CROSS_REGIONS' IN ACCOUNT
Connection: zhbcwu-tn70880

> 0 1. Yes
> 0 2. Allow OTHER in any schema
> 0 3. No
T: navigate • Enter select • 1-3 jump • Esc cancel
  
```

Screenshot: 08_CoCo_CLI_planning_prompt_first_check.png

Windows PowerShell

✓ **SQL_EXECUTE** Check Snowflake account region, edition, and parameter se...
Connection: zhsbcwu-tn708889
SQL Result (1 rows x 5 cols)

#	CURRENT_REGION()	CURRENT_ACCOUNT()	CURRENT_VERSION()	CURRENT_ROLE()	CURRENT_WAREHOUSE()
1	AMS_AP_SOUTH_1	SH96811	10.35.101	ACCOUNTADMIN	COMPUTE_WH

✓ **SQL_EXECUTE** Check cross-region Cortex inference settings
Connection: zhsbcwu-tn708889
SQL Result (1 rows x 6 cols)

#	key	value	default	level	description	type
1	CORTEX_ENABLED_CROSS_REGION	ANY_REGION	DISABLED	ACCOUNT	Comma separated list of allowed cortex cross regions. Ord...	STRING

* **SQL_EXECUTE** Check LLM completion availability in current account
Connection: zhsbcwu-tn708889
SQL compilation error:
Unknown function COMPLETE. To resolve this, an ACCOUNTADMIN must either GRANT USE AI FUNCTIONS ON ACCOUNT TO ROLE <role_name>; or GRANT USE AI FUNCTION COMPLETE ON ACCOUNT TO ROLE <role_name>; For deta...

✓ **SQL_EXECUTE** Check Cortex database roles available
Connection: zhsbcwu-tn708889
SQL Result (43 rows x 11 cols) [ctrl+t to navigate]

#	created_on	name	is_default	is_current	is_inherited	granted_to_roles	granted_to_database_roles	granted_database_roles	owner	comment
1	"2026-03-23 12:19:41.497 -0700"	AI_FUNCTIONS_USER	N	N	Y	0	0	0		
2	"2026-03-20 17:36:01.497 -0700"	AGENT_VIEWER	N	N	Y	0	0	0		
3	"2026-03-16 22:19:37.728 -0700"	ANACONDA_REPOSITORY_USER	N	N	Y	0	0	0		

Rows 1-3/43 | Cols 1-10/11

* ☐ Add Metrics ☐ Agent Model ☐ Data Gen Mode / Submit +
Add Metrics (*/*tab next, + back)
Which additional canonical supply chain metrics would you like to standardize in the semantic model?

> ☒ OTIF (On-Time In-Full) (tab to add more info)
Measures percentage of orders delivered both strictly on schedule and 100% complete with no missing lines/units.
☐ Supplier Lead Time Var
☐ Backorder Rate
☐ Inventory Turnover Ratio
☐ Something else
☐ I need clarification, let's chat about this
→ Next
? navigate * ++ switch * Enter select * Esc cancel

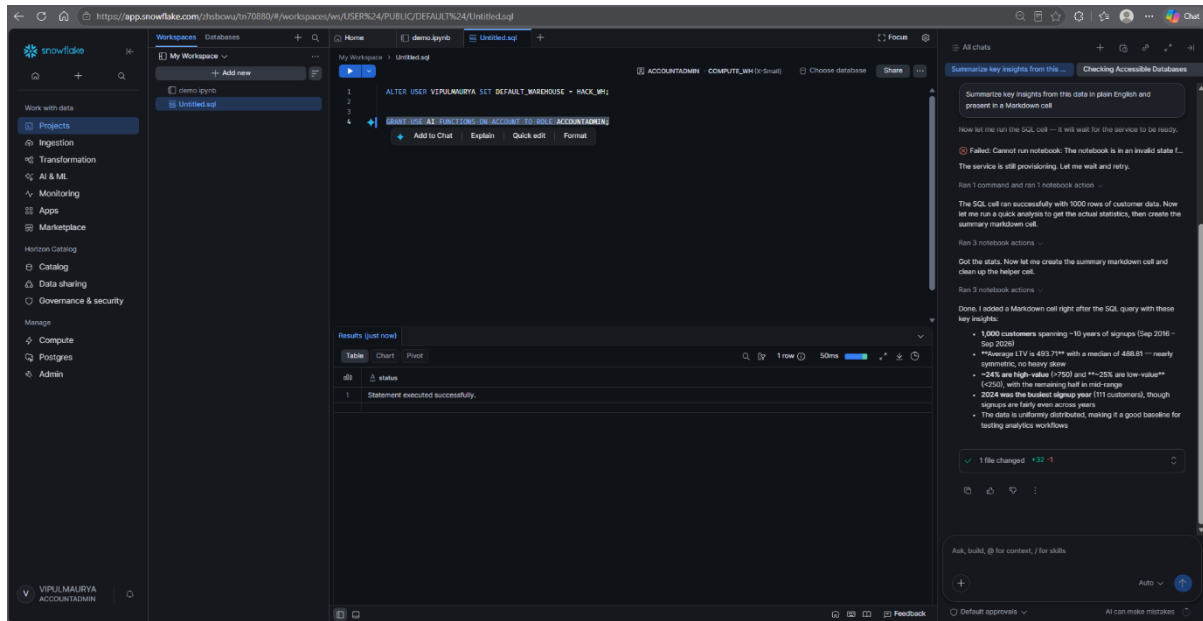
Screenshot: 09_CoCo_CLI_region_checks_AI_permission_error.png

Step 7 - Fixing the AI permission gap (Snowsight)

Date: 28 Sep 2026 | Surface: Snowsight SQL worksheet

Action: ran `GRANT USE AI FUNCTIONS ON ACCOUNT TO ROLE ACCOUNTADMIN;` (and set `DEFAULT_WAREHOUSE = HACK_WH` for the user). Result: Statement executed successfully.

Value: the missing prerequisite was discovered by CoCo during planning, not in the middle of the build - it would otherwise have blocked the Cortex Agent later.



Screenshot: 10_Snowsight_grant_AI_functions_fix.png

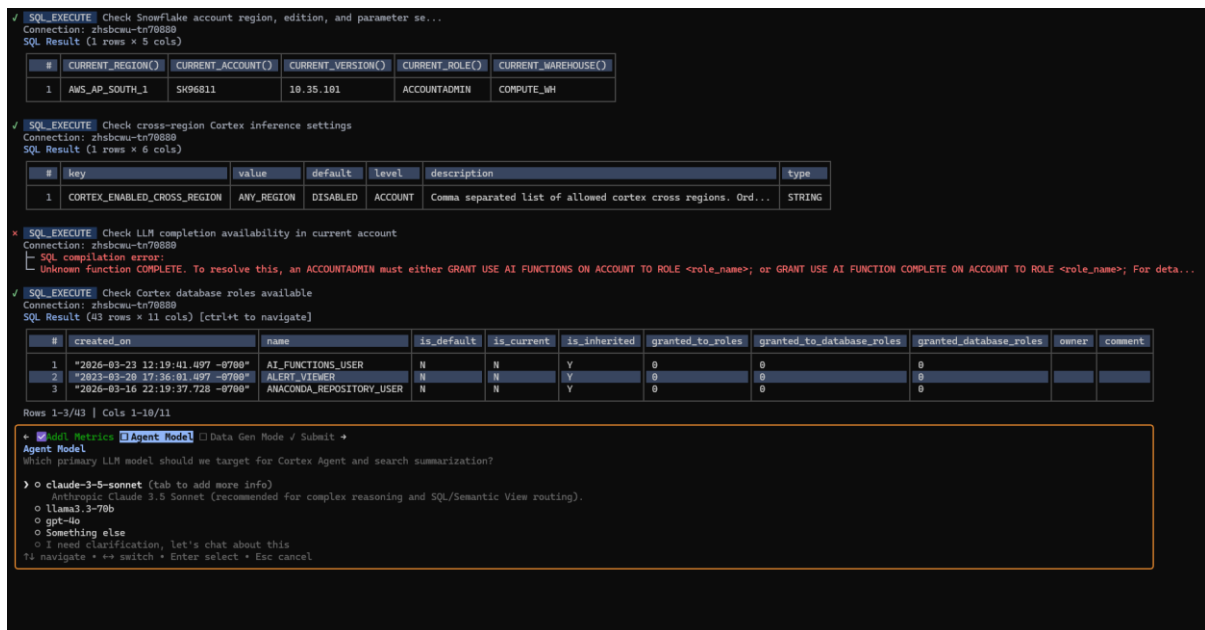
Step 8 - Answering CoCo's clarifying questions

Date: 28 Sep 2026 | Surface: CoCo CLI interactive question form

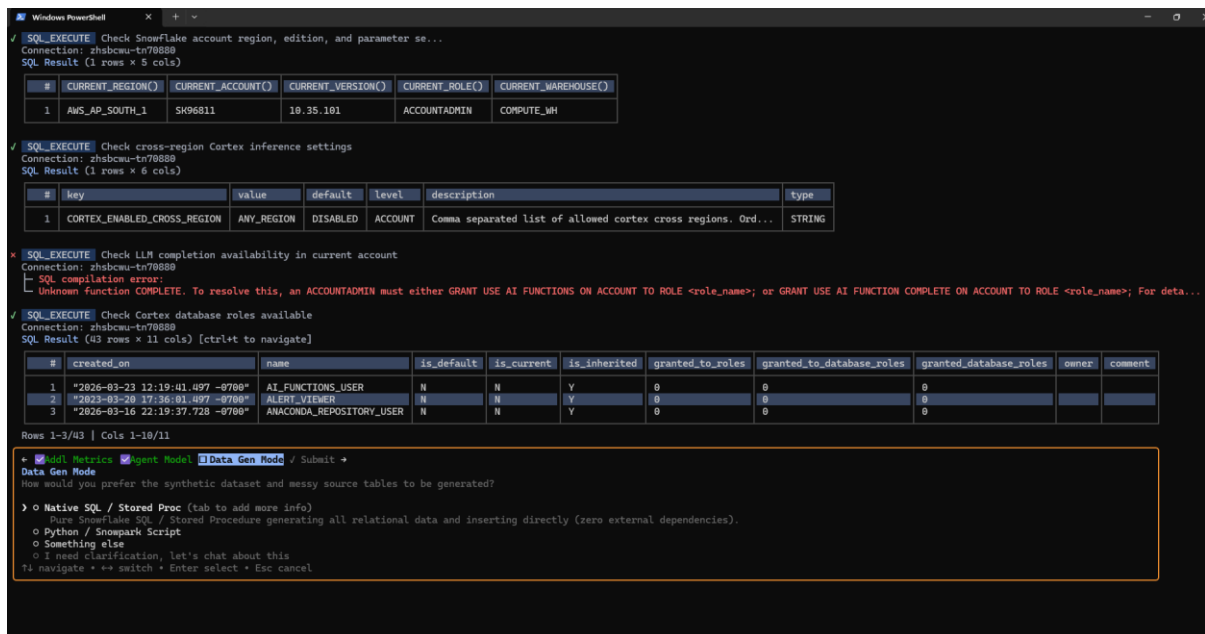
Additional metrics: selected OTIF (On-Time In-Full), Supplier Lead Time Variance and Backorder Rate (Inventory Turnover skipped - it is the inverse of Days of Inventory).

Agent model: Claude Sonnet family (strong at tool routing and SQL), available through cross-region inference.

Data generation mode: Native SQL / Stored Procedure - no external tools, cheap on an XSMALL warehouse and easy to re-run.



Screenshot: 11_CoCo_CLI_question_agent_model.png



Screenshot: 12_CoCo_CLI_question_data_generation_mode.png

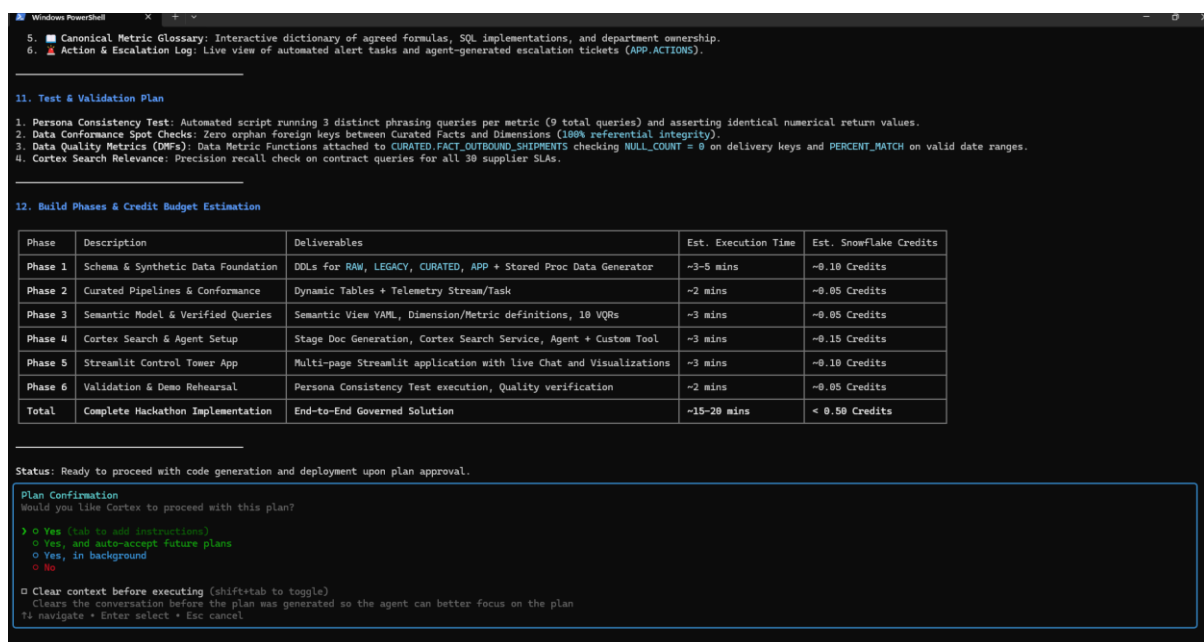
Step 9 - Reviewing CoCo's proposed plan and sending corrections

Date: 28 Sep 2026 | Surface: CoCo CLI (plan confirmation)

CoCo proposed a full blueprint (problem framing, demo story, ER diagram, RAW/LEGACY/CURATED tables, 9 metrics, data plan, pipelines, semantic view, agent, Streamlit app, tests, phases). Instead of approving blindly, the plan was reviewed and 12 corrections were sent back:

1) Cost: no task or dynamic table running every minute - dynamic tables TARGET_LAG 1 day/DOWNSTREAM, stream task hourly with SYSTEM\$STREAM_HAS_DATA, plus a daily consistency test task. 2) Use existing ACCOUNTADMIN role (HACK_ROLE did not exist). 3-4) Complete RAW sources and CURATED tables (BOM, supplier-part SLA, sales order header). 5) Landed Cost uses extended amounts and inbound freight/handling. 6) OTD/OTIF count delivered lines only. 7) Exact Days of Inventory SQL (latest snapshot / trailing 30-day consumption). 8) Late supplier renamed to Titan Micro-Foundry. 9) Demo numbers treated as targets and made consistent. 10) Contracts stored as text rows for Cortex Search. 11) Native CREATE SEMANTIC VIEW. 12) Build phase by phase with review stops.

Approval was given with the instruction: save the corrected plan as PLAN.md, create no Snowflake objects, then stop. File writes were allowed for this session only, inside the project folder.



```
5. Canonical Metric Glossary: Interactive dictionary of agreed formulas, SQL implementations, and department ownership.
6. Action & Escalation Log: Live view of automated alert tasks and agent-generated escalation tickets (APP.ACTIONS).

11. Test & Validation Plan
1. Persona Consistency Test: Automated script running 3 distinct phrasing queries per metric (9 total queries) and asserting identical numerical return values.
2. Data Conformance Spot Checks: Zero orphan foreign keys between Curated Facts and Dimensions (100% referential integrity).
3. Data Quality Metrics (DQMs): Data Metric Functions attached to CURATED.FACT_OUTBOUND.SHIPMENTS checking NULL_COUNT = 0 on delivery keys and PERCENT_MATCH on valid date ranges.
4. Cortex Search Relevance: Precision recall check on contract queries for all 30 supplier SLAs.

12. Build Phases & Credit Budget Estimation
```

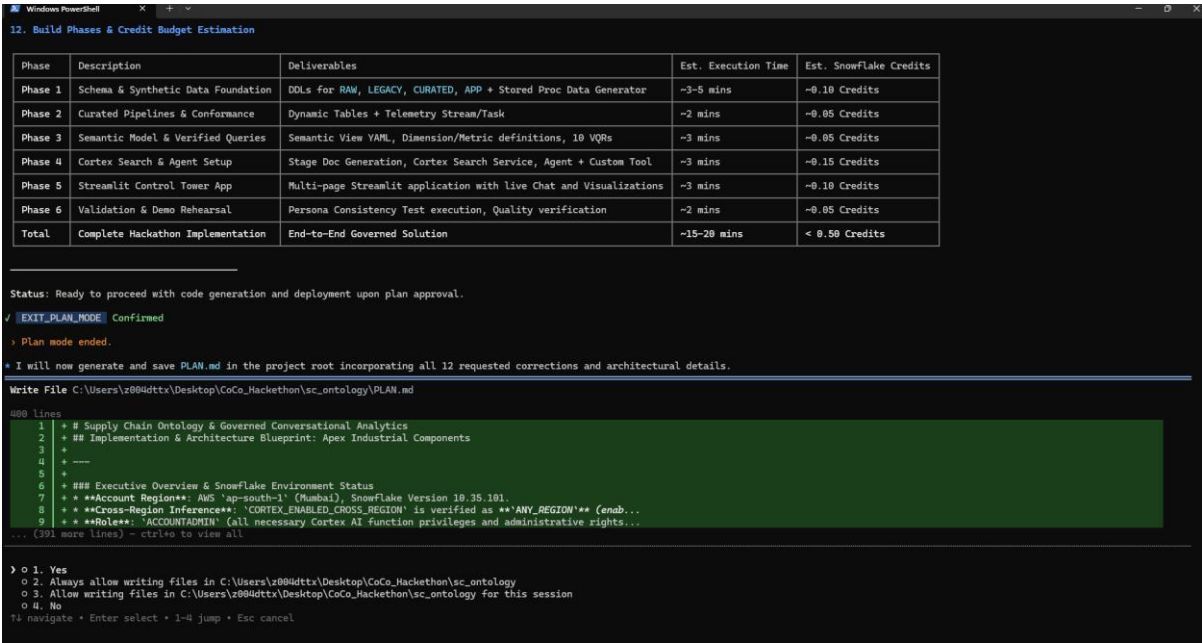
Phase	Description	Deliverables	Est. Execution Time	Est. Snowflake Credits
Phase 1	Schema & Synthetic Data Foundation	DDLs for RAW, LEGACY, CURATED, APP + Stored Proc Data Generator	~3-5 mins	~0.10 Credits
Phase 2	Curated Pipelines & Conformance	Dynamic Tables + Telemetry Stream/Task	~2 mins	~0.05 Credits
Phase 3	Semantic Model & Verified Queries	Semantic View YAML, Dimension/Metric definitions, 10 VQRs	~3 mins	~0.05 Credits
Phase 4	Cortex Search & Agent Setup	Stage Doc Generation, Cortex Search Service, Agent + Custom Tool	~3 mins	~0.15 Credits
Phase 5	Streamlit Control Tower App	Multi-page Streamlit application with Live Chat and Visualizations	~3 mins	~0.10 Credits
Phase 6	Validation & Demo Rehearsal	Persona Consistency Test execution, Quality verification	~2 mins	~0.05 Credits
Total	Complete Hackathon Implementation	End-to-End Governed Solution	~15-20 mins	< 0.50 Credits

```
Status: Ready to proceed with code generation and deployment upon plan approval.

Plan Confirmation
Would you like Cortex to proceed with this plan?
> ☐ Yes (tab to add instructions)
   ☐ Yes, and auto-accept future plans
   ☐ Yes, in background
   ☐ No

☐ Clear context before executing (shift+tab to toggle)
Clears the conversation before the plan was generated so the agent can better focus on the plan
?4 navigate • Enter select • Esc cancel
```

Screenshot: 13_CoCo_CLI_proposed_plan_confirmation.png



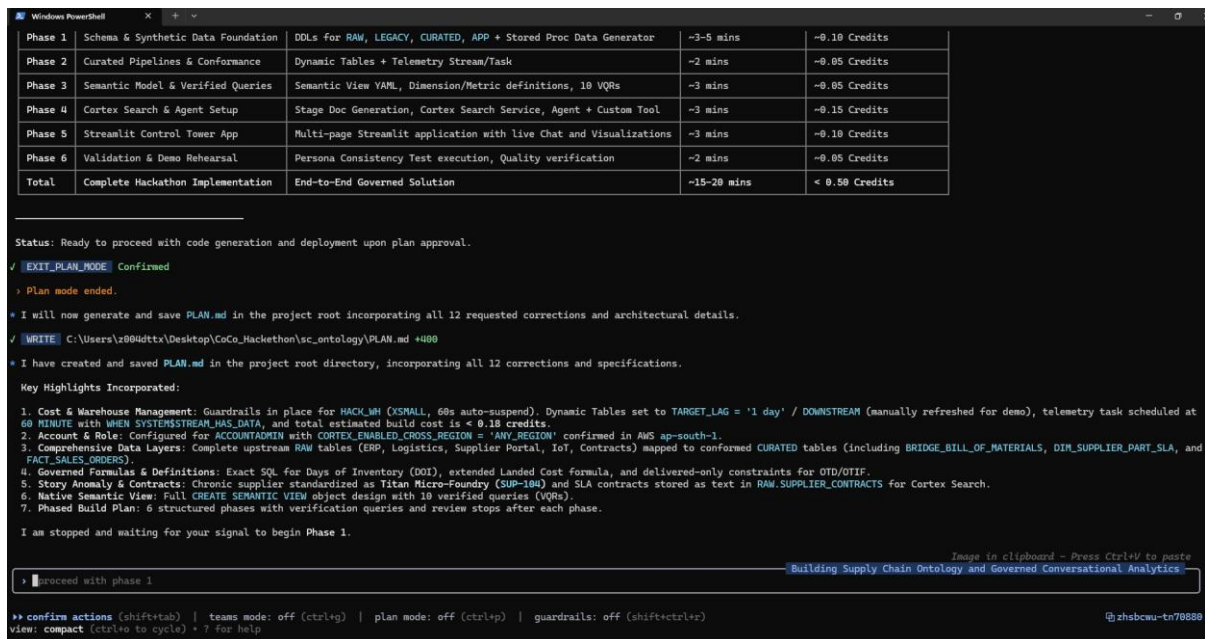
Screenshot: 14_CoCo_CLI_writing_PLAN_md.png

Step 10 - PLAN.md saved (end of Planning phase)

Date: 28 Sep 2026 | Output: sc_ontology\PLAN.md (400 lines) + CoCo's own plan copy in sc_ontology\.cortex\plans\

CoCo confirmed all 12 corrections: cost guardrails (estimated build compute < 0.18 credits), ACCOUNTADMIN role, complete RAW-to-CURATED lineage incl. BRIDGE_BILL_OF_MATERIALS, DIM_SUPPLIER_PART_SLA and FACT_SALES_ORDERS, governed formulas, Titan Micro-Foundry (SUP-104) story anomaly, SLA contracts in RAW.SUPPLIER_CONTRACTS, native semantic view with 10 verified queries, and 6 build phases with verification stops.

Status: no Snowflake objects created yet; CoCo is stopped and waiting for approval to start Phase 1 (database, schemas and synthetic data).



Phase	Task	DDLs for RAW, LEGACY, CURATED, APP + Stored Proc Data Generator	Time	Credits
Phase 1	Schema & Synthetic Data Foundation	DDLs for RAW, LEGACY, CURATED, APP + Stored Proc Data Generator	~3-5 mins	~0.10 Credits
Phase 2	Curated Pipelines & Conformance	Dynamic Tables + Telemetry Stream/Task	~2 mins	~0.05 Credits
Phase 3	Semantic Model & Verified Queries	Semantic View YAML, Dimension/Metric definitions, 10 VQRs	~3 mins	~0.05 Credits
Phase 4	Cortex Search & Agent Setup	Stage Doc Generation, Cortex Search Service, Agent + Custom Tool	~3 mins	~0.15 Credits
Phase 5	Streamlit Control Tower App	Multi-page Streamlit application with Live Chat and Visualizations	~3 mins	~0.10 Credits
Phase 6	Validation & Demo Rehearsal	Persona Consistency Test execution, Quality verification	~2 mins	~0.05 Credits
Total	Complete Hackathon Implementation	End-to-End Governed Solution	~15-20 mins	< 0.50 Credits

Status: Ready to proceed with code generation and deployment upon plan approval.

✓ EXIT_PLAN_MODE Confirmed

Plan mode ended.

I will now generate and save PLAN.md in the project root incorporating all 12 requested corrections and architectural details.

✓ WRITE C:\Users\r894dtt\Desktop\CoCo_Hackathon\sc_ontology\PLAN.md #400

I have created and saved PLAN.md in the project root directory, incorporating all 12 corrections and specifications.

Key Highlights Incorporated:

1. Cost & Warehouse Management: Guardrails in place for HACK_MH (XSMALL, 60s auto-suspend). Dynamic Tables set to TARGET_LAG = '1 day' / DOWNSTREAM (manually refreshed for demo), telemetry task scheduled at 60 MINUTE with WHEN SYSTEMSTREAM_HAS_DATA, and total estimated build cost is < 0.18 credits.
2. Account & Role: Configured for ACCOUNTADMIN with CORTEX_ENABLED_CROSS_REGION = 'ANY_REGION' confirmed in AWS ap-south-1.
3. Comprehensive Data Layers: Complete upstream RAW tables (ERP, Logistics, Supplier Portal, IoT, Contracts) mapped to conformed CURATED tables (including BRIDGE_BILL_OF_MATERIALS, DIM_SUPPLIER_PART_SLA, and FACT_SALES_ORDERS).
4. Governed Formulas & Definitions: Exact SQL for Days of Inventory (DOI), extended Landed Cost formula, and delivered-only constraints for OTD/OTIF.
5. Story Anomaly & Contracts: Chronic supplier standardized as Titan Micro-Foundry (SUP-104) and SLA contracts stored as text in RAW.SUPPLIER_CONTRACTS for Cortex Search.
6. Native Semantic View: Full CREATE SEMANTIC VIEW object design with 10 verified queries (VQRs).
7. Phased Build Plan: 6 structured phases with verification queries and review stops after each phase.

I am stopped and waiting for your signal to begin Phase 1.

Image in clipboard - Press Ctrl+V to paste

Building Supply Chain Ontology and Governed Conversational Analytics

proceed with phase 1

confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)

view: compact (ctrl+o to cycle) * 7 for help

@zhbcsu-tn70889

Screenshot: 15_CoCo_CLI_PLAN_md_saved_waiting_phase1.png

Step 11 - Phase 1 build: database, schemas and RAW source tables

Date: 28 Sep 2026 | Surface: CoCo CLI | Warehouse: HACK_WH (XSMALL, 60 s auto-suspend)

Prompt (summary): proceed with Phase 1 only, following PLAN.md; reuse HACK_WH; data range 1 Oct 2025 - 30 Sep 2026 so Q3 2026 is the demo quarter; add RAW.FX_RATES and RAW.UOM_CONVERSIONS; deterministic, re-runnable generator; create the 3 LEGACY views; save every script under sql/; verify the demo patterns and stop.

Result: CoCo created database SC_ONTOLOGY with schemas RAW, LEGACY, CURATED, SEMANTIC and APP, 19 RAW tables simulating ERP, logistics, supplier portal, IoT and contract sources, and the stored procedure RAW.SP_GENERATE_SYNTHETIC_SUPPLY_CHAIN_DATA. Scripts saved locally: 01_setup.sql, 02_raw_tables.sql, 03_generate_data.sql, 04_legacy_views.sql. Each SQL statement was reviewed before approval; generated scripts were checked to touch only SC_ONTOLOGY.

The screenshot displays the CoCo CLI interface. At the top, a table summarizes the project phases:

Phase	Task	Duration	Credits	
Phase 2	Curated Pipelines & Conformance	Dynamic Tables + Telemetry Stream/Task	~2 mins	~0.05 Credits
Phase 3	Semantic Model & Verified Queries	Semantic View YAML, Dimension/Metric definitions, 10 VQRs	~3 mins	~0.05 Credits
Phase 4	Cortex Search & Agent Setup	Stage Doc Generation, Cortex Search Service, Agent + Custom Tool	~3 mins	~0.15 Credits
Phase 5	Streamlit Control Tower App	Multi-page Streamlit application with live Chat and Visualizations	~3 mins	~0.10 Credits
Phase 6	Validation & Demo Rehearsal	Persona Consistency Test execution, Quality verification	~2 mins	~0.05 Credits
Total	Complete Hackathon Implementation	End-to-End Governed Solution	~15-20 mins	< 0.50 Credits

Below the table, the CLI shows the status: "Ready to proceed with code generation and deployment upon plan approval." It confirms the exit of plan mode and generates a new PLAN.md file. Key highlights include:

- 1. Cost & Warehouse Management: Guardrails in place for HACK_WH (XSMALL, 60s auto-suspend). Dynamic Tables set to TARGET_LAG = '1 day' / DOWNSTREAM (manually refreshed for demo), telemetry task scheduled at 60 MINUTE with WHEN SYSTEMSTREAM_JAS_DATA, and total estimated build cost is < 0.10 credits.
- 2. Account & Role: Configured for ACCOUNTADMIN with CORTEX_ENABLED_CROSS_REGION = 'ANY_REGION' confirmed in AWS ap-south-1.
- 3. Comprehensive Data Layers: Complete upstream RAW tables (ERP, Logistics, Supplier Portal, IoT, Contracts) mapped to conformed CURATED tables (including BRIDGE_BILL_OF_MATERIALS, DIM_SUPPLIER_PART_SLA, and FACT_SALES_ORDERS).
- 4. Governed Formulas & Definitions: Exact SQL for Days of Inventory (DOI), extended Landed Cost formula, and delivered-only constraints for OTD/OTIF.
- 5. Story Anomaly & Contracts: Chronic supplier standardized as Titan Micro-Foundry (SUP-100) and SLA contracts stored as text in RAW.SUPPLIER_CONTRACTS for Cortex Search.
- 6. Native Semantic View: Full CREATE SEMANTIC VIEW object design with 10 verified queries (VQRs).
- 7. Phased Build Plan: 6 structured phases with verification queries and review steps after each phase.

The CLI then prompts to proceed with Phase 1, which is building the Supply Chain Ontology and Governed Conversational Analytics.

Screenshot: 16_CoCo_CLI_phase1_prompt_create_warehouse.png

```
Windows PowerShell
- PART-KC-02 stockout days at US-TX-01 in August
- CARR-04 outbound OTD in Jul-Aug (target about 58%)
- Q3 2026 OTD from each of the 3 LEGACY views (they must differ)
Then STOP and wait for my review.

* I will now begin Phase 1 execution by setting up the database, schemas, warehouse, RAW tables, deterministic data generator, legacy views, and local SQL script files.

✓ WRITE C:\Users\z004dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\01_setup.sql +37
✓ SQL_EXECUTE Execute database, schema, and warehouse setup
Connection: zhsbcwu-tr70880
SQL Result (1 rows x 1 cols)


| # | status                           |
|---|----------------------------------|
| 1 | Schema APP successfully created. |



✓ WRITE C:\Users\z004dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\02_raw_tables.sql +194
✓ SQL_EXECUTE Create all RAW tables and reference tables
Connection: zhsbcwu-tr70880
SQL Result (1 rows x 1 cols)


| # | status                                         |
|---|------------------------------------------------|
| 1 | Table SUPPLIER_CONTRACTS successfully created. |



✓ WRITE C:\Users\z004dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\03_generate_data.sql +370
✓ SQL_EXECUTE Deploy synthetic data generator stored procedure
Connection: zhsbcwu-tr70880
SQL Result (1 rows x 1 cols)


| # | status                                                       |
|---|--------------------------------------------------------------|
| 1 | Function SP_GENERATE_SYNTHETIC_SUPPLY_CHAIN_DATA successf... |



Execute SQL
CALL RAW.SP_GENERATE_SYNTHETIC_SUPPLY_CHAIN_DATA()
Connection: zhsbcwu-tr70880

> o 1. Yes
o 2. Allow OTHER in any schema
o 3. No
? navigate * Enter select * 1-3 jump * Esc cancel
Tip: Use /guardrails to set a Restricted Session Scope and bypass future SQL permission prompts.
```

Screenshot: 17_CoCo_CLI_phase1_schemas_raw_tables_generator.png

Step 12 - Synthetic data: self-correction and calibration

Date: 28 Sep 2026 | Surface: CoCo CLI

Error handling: the first generator run failed ('duplicate alias values'); a later calibrated version failed with a syntax error. In both cases CoCo diagnosed the error, rewrote 03_generate_data.sql and re-ran it without manual code changes.

Self-validation: after the first successful load CoCo compared the planted story patterns with the targets and found them too weak (SUP-104 OTIF 73.68% vs ~64%, overrun +2.39 vs +14 days, CARR-04 OTD 100% vs ~58%). It recalibrated the generator on its own.

Human review: the legacy Planning view then returned exactly 100% on-time for Q3 2026 - unrealistic for the 'before' story. A follow-up prompt asked CoCo to make ~15% of shipments dispatch late (target 81-85%) while keeping the other anomalies; CoCo updated the generator and re-ran it.

```

SQL Result (1 rows x 1 cols)
# | status
1 | Table SUPPLIER_CONTRACTS successfully created.

✓ WRITE C:\Users\z804dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\03_generate_data.sql +370

✓ SQL_EXECUTE Deploy synthetic data generator stored procedure
Connection: zhsbcwu-tr78889
SQL Result (1 rows x 1 cols)
# | status
1 | Function SP_GENERATE_SYNTHETIC_SUPPLY_CHAIN_DATA successf...

✗ SQL_EXECUTE Execute synthetic data generation procedure
Connection: zhsbcwu-tr78889
└─ Uncaught exception of type 'STATEMENT_ERROR' on line 23 at position 4 : SQL compilation error:
   duplicate alias 'values'

✓ WRITE C:\Users\z804dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\03_generate_data.sql +373 -370

✓ SQL_EXECUTE Recreate and execute synthetic data generator procedure
Connection: zhsbcwu-tr78889
SQL Result (1 rows x 1 cols)
# | SP_GENERATE_SYNTHETIC_SUPPLY_CHAIN_DATA
1 | SUCCESS: Deterministic synthetic data generated for 1 Oct...

✓ WRITE C:\Users\z804dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\04_legacy_views.sql +72

✓ SQL_EXECUTE Deploy legacy reporting views
Connection: zhsbcwu-tr78889
SQL Result (1 rows x 1 cols)
# | status
1 | View VM_LOGISTICS_OTD successfully created.

✓ SQL_EXECUTE Query row counts for all RAW tables
Connection: zhsbcwu-tr78889
SQL Result (19 rows x 2 cols) [ctrl+t to navigate]
# | TABLE_NAME | ROW_COUNT

```

Screenshot: 18_CoCo_CLI_phase1_selfcorrect_and_first_validation.png

```
Windows PowerShell
Rows 1-3/19

✓ SQL_EXECUTE Verify Titan Micro-Foundry (SUP-104) Inbound OTIF and Lea...
Connection: zhsbcwu-tn70888
SQL Result (1 rows x 8 cols)

# | SUPPLIER_CODE | VENDOR_NAME | TOTAL_RECEIVED_LINES | ON_TIME_LINES | IN_FULL_LINES | OTIF_LINES | INBOUND_OTIF_PCT | AVG_LEAD_TIME_OVERRUN_DAYS |
--|---|---|---|---|---|---|---|---|
1 | SUP-104 | Titan Micro-Foundry | 494 | 364 | 494 | 364 | 73.68 | 2.39 |

✓ SQL_EXECUTE Check PART-MC-02 stockout days at US-TX-01 in August 2026
Connection: zhsbcwu-tn70888
SQL Result (8 rows x 4 cols) [ctrl+t to navigate]

# | PLANT_CODE | MATERIAL_NO | SNAP_DATE_STR | STOCK_ON_HAND |
--|---|---|---|---|
1 | US-TX-01 | PART-MC-02 | 2026-08-08 | 564.00 |
2 | US-TX-01 | PART-MC-02 | 2026-08-09 | 974.00 |
3 | US-TX-01 | PART-MC-02 | 2026-08-10 | 0.00 |

Rows 1-3/8

✓ SQL_EXECUTE Check CARR-B4 Outbound OTD in Jul-Aug 2026
Connection: zhsbcwu-tn70888
SQL Result (1 rows x 5 cols)

# | CARRIER_ID | CARRIER_NAME | TOTAL_JUL_AUG_DELIVERIES | ON_TIME_DELIVERIES | OUTBOUND_OTD_PCT |
--|---|---|---|---|---|
1 | CARR-B4 | Pacific Express Ocean Lines | 214 | 214 | 100.00 |

✓ WRITE C:\Users\z004dtt\Desktop\CoCo_Hackathon\sc_ontology\sql\03-generate_data.sql +374 -373

* SQL_EXECUTE Recreate and run calibrated synthetic data generator
Connection: zhsbcwu-tn70888
- SQL compilation error:
- syntax error line 10 at position 13 unexpected 'TABLE'.
- syntax error line 10 at position 22 unexpected '.,'.

F Waddling... (esc to interrupt)
  Tip: Ask "Create a Streamlit app" to build an interactive dashboard

Image in clipboard - Press Ctrl+V to paste
Building Supply Chain Ontology and Governed Conversational Analytics

Type to queue message...

confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+o to cycle) * Type to queue next message * esc to interrupt @zhsbcwu-tn70888
```

Screenshot: 19_CoCo_CLI_phase1_anomalies_off_target_recalibrate.png

```
Windows PowerShell

OUTBOUND_OTD_PCT
FROM RAW.LOGISTICS_SHIPMENTS s
JOIN RAW.LOGISTICS_CARRIERS c ON s.CARRIER_ID = c.CARRIER_CODE
JOIN RAW.ERP_SALES_ORDERS so ON s.SO_HEADER_REF = so.SO_NUMBER
WHERE s.CARRIER_ID = 'CARR-B4' AND s.STATUS = 'DELIVERED'
AND TRY_TO_DATE(so.SO_DATE_STR, 'YYYY-MM-DD') BETWEEN '2026-07-01' AND '2026-08-31'
GROUP BY 1, 2;

* Result: 60.93% Outbound OTD (131 on-time deliveries out of 215 shipments, demonstrating severe monsoon delays).

5. Q3 2026 OTD Across 3 Legacy Views (The "Before" Conflict)

SELECT
  'Planning View (Plant Dispatch <= Promised Date)' AS LEGACY_VIEW,
  SUM(TOTAL_SHIPMENTS) AS TOTAL_RECORDS,
  ROUND(100.0 * SUM(ON_TIME_DISPATCH_COUNT) / SUM(TOTAL_SHIPMENTS), 2) AS OVERALL_Q3_OTD_PCT
FROM LEGACY_VW_PLANNING_OTD WHERE QUARTER_LABEL = 'Q3 2026'
UNION ALL
SELECT
  'Procurement View (GR Date <= Supplier Commit + 3d Grace)' AS LEGACY_VIEW,
  SUM(TOTAL_RECEIPTS) AS TOTAL_RECORDS,
  ROUND(100.0 * SUM(ON_TIME_GR_COUNT) / SUM(TOTAL_RECEIPTS), 2) AS OVERALL_Q3_OTD_PCT
FROM LEGACY_VW_PROCUREMENT_OTD WHERE QUARTER_LABEL = 'Q3 2026'
UNION ALL
SELECT
  'Logistics View (Actual POD <= Customer Requested Date)' AS LEGACY_VIEW,
  SUM(TOTAL_DELIVERIES) AS TOTAL_RECORDS,
  ROUND(100.0 * SUM(ON_TIME_DELIVERY_COUNT) / SUM(TOTAL_DELIVERIES), 2) AS OVERALL_Q3_OTD_PCT
FROM LEGACY_VW_LOGISTICS_OTD WHERE QUARTER_LABEL = 'Q3 2026';

* Result:
* Planning View: 100.00%
* Procurement View: 96.65%
* Logistics View: 76.14%

The 3 legacy perspectives return three completely conflicting numbers for Q3 2026 delivery performance.

Status: Stopped and standing by for your review before proceeding to Phase 2.

Proceed with Phase 2

Building Supply Chain Ontology and Governed Conversational Analytics

confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+o to cycle) * ? for help @zhsbcwu-tn70888
```

Screenshot: 20_CoCo_CLI_phase1_legacy_views_planning_100pct.png

Step 13 - Phase 1 final verification (all targets met)

Date: 28 Sep 2026 | Surface: CoCo CLI

Row counts: 4 plants, 30 suppliers, 200 parts, 50 products, 100 customers, 8 carriers, 250 BOM rows; 5,000 purchase orders, 5,000 goods receipts, 5,000 sales orders, 15,000 order lines, 5,000 shipments, 15,000 shipment lines; 73,000 inventory snapshots, 2,500 IoT events, 30 supplier contracts, 36 FX rates, 6 UOM conversions.

Story patterns: Titan Micro-Foundry (SUP-104) inbound OTIF 64.00% with +14.00 days lead-time overrun; PART-MC-02 stockout at US-TX-01 for 4 days (10-13 Aug 2026); Pacific Express (CARR-04) outbound OTD 54.42% in Jul-Aug monsoon season.

'Before' problem proven: for Q3 2026 the three legacy team views report 83.51% (Planning: dispatch vs promised date), 96.65% (Procurement: goods receipt vs supplier commit + 3-day grace) and 64.12% (Logistics: proof of delivery vs customer requested date) - three conflicting answers to 'What is our on-time delivery?'.

```
Windows PowerShell
Connection: zhshcwu-tn78888
SQL Result (3 rows x 4 cols)

# LEGACY_VIEW | TOTAL_RECORDS | ON_TIME_RECORDS | OVERALL_Q3_OTD_PCT
1 Planning View (Plant Dispatch <= Promised Date) | 1192 | 962 | 83.51
2 Procurement View (GR Date <= Supplier Commit + 3d Grace) | 1194 | 1154 | 96.65
3 Logistics View (Actual POD <= Customer Requested Date) | 1198 | 731 | 64.12

/ SQL_EXECUTE Verify SUP-104 Inbound OTIF and lead time overrun
Connection: zhshcwu-tn78888
SQL Result (3 rows x 6 cols)

# SUPPLIER_CODE | VENDOR_NAME | TOTAL_RECEIVED_LINES | OTIF_LINES | INBOUND_OTIF_PCT | AVG_DELAYED_OVERRUN_DAYS
1 SUP-104 | Titan Micro-Foundry | 500 | 320 | 64.00 | 14.00

/ SQL_EXECUTE Verify CARR-04 outbound OTD in Jul-Aug 2026
Connection: zhshcwu-tn78888
SQL Result (1 rows x 5 cols)

# CARRIER_ID | CARRIER_NAME | TOTAL_JUL_AUG_DELIVERIES | ON_TIME_DELIVERIES | OUTBOUND_OTD_PCT
1 CARR-04 | Pacific Express Ocean Lines | 215 | 117 | 54.42

Phase 1 Compact Verification Summary

Category / Verification Item | Key Metric / Detail | Verified Value | Target / Status
RAW Layer Master Row Counts | PLANTS / SUPPLIERS / PARTS / PRODUCTS / CUSTOMERS / CARRIERS / BOM | 4 / 30 / 200 / 50 / 100 / 8 / 250 | Complete
RAW Layer Transaction Row Counts | PURCHASE_ORDERS / GOODS_RECEIPTS / SALES_ORDERS / SO_LINES / SHIPMENTS / SHIP_LINES | 5,000 / 5,000 / 5,000 / 15,000 / 5,000 / 15,000 | Complete
RAW Layer Snapshots, Telemetry & Ref | INVENTORY_SNAPSHOTS / IOT_TELEMETRY / CONTRACTS / FX_RATES / UOM_CONV | 73,000 / 2,500 / 30 / 36 / 6 | Complete
Supplier SUP-104 (Titan Micro-Foundry) | Inbound OTIF Rate & Avg Lead-Time Overrun on Delayed Lines | 64.00% (320/500 lines) & +14.00 days | Matches ~64% target & +14d overrun
Component PART-MC-02 August Stockout | Stockout Days (STOCK_ON_HAND = 0.00) at Plant US-TX-01 in Aug 2026 | 4 days (2026-08-10 through 2026-08-13) | Matches target
Carrier CARR-04 (Pacific Express) | Outbound OTD in Monsoon Period (Jul-Aug 2026) | 54.42% (117/215 shipments on-time) | Matches ~55% delay target
Q3 2026 Legacy View: Planning | LEGACY_VW_PLANNING_OTD (Plant Dispatch Date vs Promised Date) | 83.51% (962/1,152 shipments) | Matches 81-85% target
Q3 2026 Legacy View: Procurement | LEGACY_VW_PROCUREMENT_OTD (Goods Receipt vs Supplier Commit + 3d Grace) | 96.65% (1,154/1,194 lines) | Demonstrates conflicting view
Q3 2026 Legacy View: Logistics | LEGACY_VW_LOGISTICS_OTD (Actual POD Delivery vs Customer Requested Date) | 64.12% (731/1,148 deliveries) | Demonstrates conflicting view

Updated file: sql/03_generate_data.sql.

Status: Phase 1 is complete. Stopped and waiting for your review.

[Proceed with Phase 2.] Building Supply Chain Ontology and Governed Conversational Analytics

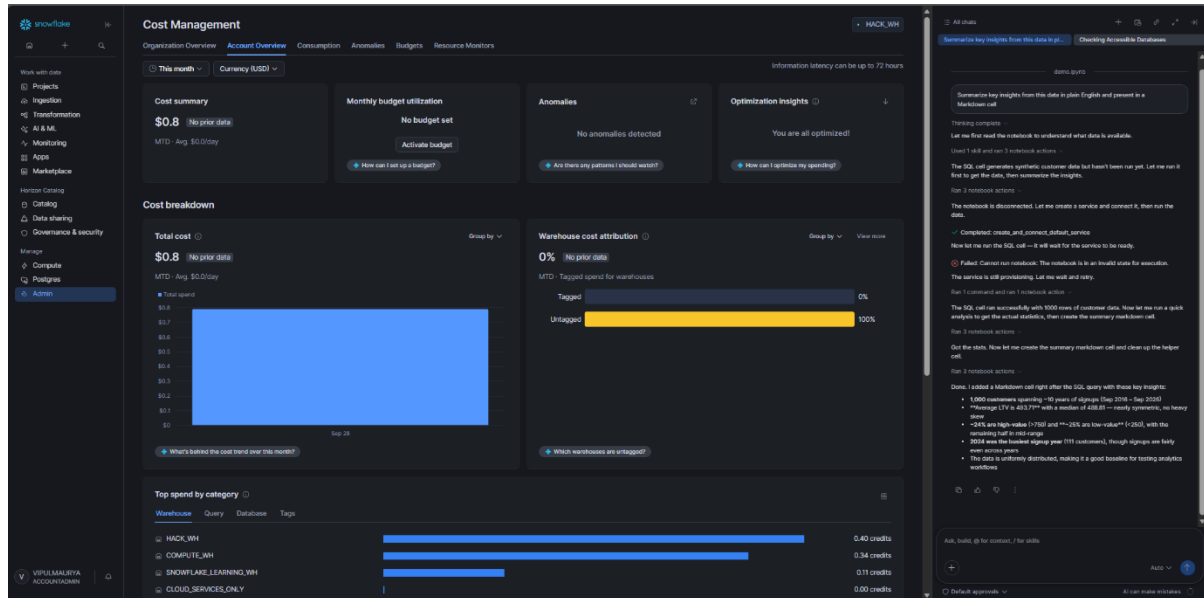
>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r) @ zhshcwu-tn78888
view: compact (ctrl+to cycle) + ? for help
```

Screenshot: 21_CoCo_CLI_phase1_final_verification_summary.png

Step 14 - Cost check after Phase 1 (Snowsight)

Date: 28 Sep 2026 | Surface: Snowsight > Admin > Cost Management > Account Overview

Result: about USD 0.8 spent this month (HACK_WH 0.40, COMPUTE_WH 0.34, SNOWFLAKE_LEARNING_WH 0.11 credits) out of the USD 400 trial balance. Cost guardrails in place: XSMALL warehouse, 60-second auto-suspend, no always-on tasks.



Screenshot: 22_Snowsight_cost_management_after_phase1.png

Step 15 - Phase 2 curated pipelines and quality checks

Date: 28 Sep 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH (XSMALL)

Prompt summary: build the conformed CURATED dynamic tables, vendor cross-reference and FX/UOM normalization, current inventory position for DOI, APP action and consistency logs, and the telemetry stream with two scheduled tasks left suspended. Save the SQL scripts and run the Phase 2 checks.

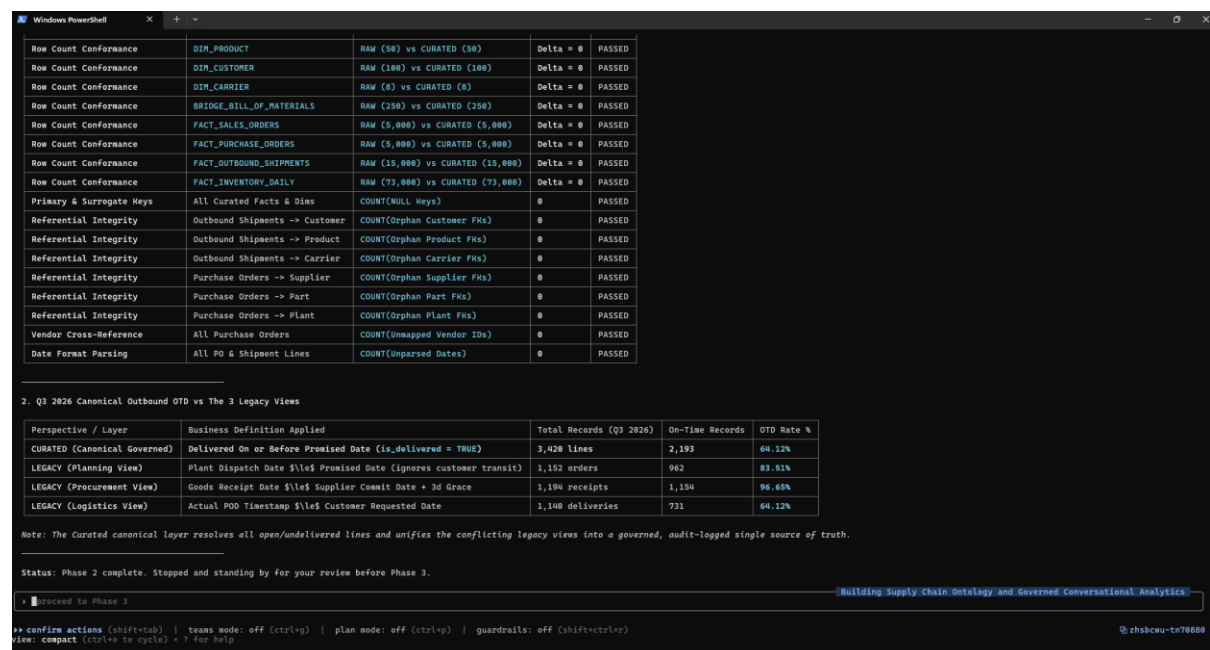
Result: Phase 2 completed. Scripts saved as sql/05_curated_dynamic_tables.sql, sql/06_streams_tasks.sql and sql/07_quality_checks.sql. CoCo reported all displayed row-count conformance checks with Delta = 0 and PASSED. Null/surrogate key, outbound and purchase-order foreign-key, vendor mapping, and date parsing checks shown in the summary each returned 0 and PASSED.

Step 16 - Q3 2026 canonical OTD comparison

CoCo reported Curated canonical Outbound OTD of 64.12% (2,193 on-time lines out of 3,420) against Legacy Planning 83.51% (962/1,152), Legacy Procurement 96.65% (1,154/1,194), and Legacy Logistics 64.12% (731/1,140). The governed definition shown was delivered on or before promised date, with is_delivered = TRUE.

Review note for Phase 3: canonical and Legacy Logistics have the same displayed rate. The line counts and on-time counts are each exactly three times the Logistics delivery counts, so verify the promised-date versus requested-date logic and the shipment-line versus delivery grain before presenting this as independent proof of metric convergence.

Screenshot: 23_CoCo_CLI_phase2_curated_quality_and_q3_otd.png



The screenshot displays the CoCo CLI interface in a Windows PowerShell window. It shows the results of Phase 2 quality checks and a comparison of Q3 2026 canonical OTD against legacy views.

Check Type	Check Name	Raw Count	Curated Count	Delta	Status
Row Count Conformance	DIH_PRODUCT	RAW (50) vs CURATED (50)		Delta = 0	PASSED
Row Count Conformance	DIH_CUSTOMER	RAW (100) vs CURATED (100)		Delta = 0	PASSED
Row Count Conformance	DIH_CARRIER	RAW (8) vs CURATED (8)		Delta = 0	PASSED
Row Count Conformance	BRIDGE_BILL_OF_MATERIALS	RAW (250) vs CURATED (250)		Delta = 0	PASSED
Row Count Conformance	FACT_SALES_ORDERS	RAW (5,000) vs CURATED (5,000)		Delta = 0	PASSED
Row Count Conformance	FACT_PURCHASE_ORDERS	RAW (5,000) vs CURATED (5,000)		Delta = 0	PASSED
Row Count Conformance	FACT_OUTBOUND_SHIPMENTS	RAW (15,000) vs CURATED (15,000)		Delta = 0	PASSED
Row Count Conformance	FACT_INVENTORY_DAILY	RAW (75,000) vs CURATED (75,000)		Delta = 0	PASSED
Primary & Surrogate Keys	All Curated Facts & Dims	COUNT(NULL Keys)	0		PASSED
Referential Integrity	Outbound Shipments -> Customer	COUNT(Orphan Customer FKs)	0		PASSED
Referential Integrity	Outbound Shipments -> Product	COUNT(Orphan Product FKs)	0		PASSED
Referential Integrity	Outbound Shipments -> Carrier	COUNT(Orphan Carrier FKs)	0		PASSED
Referential Integrity	Purchase Orders -> Supplier	COUNT(Orphan Supplier FKs)	0		PASSED
Referential Integrity	Purchase Orders -> Part	COUNT(Orphan Part FKs)	0		PASSED
Referential Integrity	Purchase Orders -> Plant	COUNT(Orphan Plant FKs)	0		PASSED
Vendor Cross-Reference	All Purchase Orders	COUNT(Unmapped Vendor IDs)	0		PASSED
Date Format Parsing	All PO & Shipment Lines	COUNT(Unparsed Dates)	0		PASSED

Perspective / Layer	Business Definition Applied	Total Records (Q3 2026)	On-Time Records	OTD Rate %
CURATED (Canonical Governed)	Delivered On or Before Promised Date (is_delivered = TRUE)	3,420 Lines	2,193	64.12%
LEGACY (Planning View)	Plant Dispatch Date \$!le\$ Promised Date (ignores customer transit)	1,152 orders	962	83.51%
LEGACY (Procurement View)	Goods Receipt Date \$!le\$ Supplier Commit Date + 3d Grace	1,194 receipts	1,154	96.65%
LEGACY (Logistics View)	Actual POD Timestamp \$!le\$ Customer Requested Date	1,140 deliveries	731	64.12%

Note: The Curated canonical layer resolves all open/undelivered lines and unifies the conflicting legacy views into a governed, audit-logged single source of truth.

Status: Phase 2 complete. Stopped and standing by for your review before Phase 3.

Building Supply Chain Ontology and Governed Conversational Analytics

Proceed to Phase 3

confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+u to cycle) | ? for help

@zhsbcwu-tn70880

Step 17 - Phase 3 semantic model and validation

Date: 28 Sep 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH

Result: Native semantic view SC_ontology.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE and 10 AI_VERIFIED_QUERIES saved in sql/08_semantic_view.sql; query tests saved in sql/09_verified_queries.sql. Phase 3.1 corrected DOI grouped aggregation to total on-hand divided by total daily consumption and limited delayed-line counts to delivered lines. The three Phase 3 persona-labeled SQL rows were a SQL parity check, not natural-language Analyst validation.

Step 18 - Phase 4 Search and Agent setup (validation partial)

Date: 28 Sep 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH

CoCo confirmed the Search service and Agent were absent, checked APP.SP_CREATE_ACTION_ITEM grants (OWNERSHIP to ACCOUNTADMIN), then created CSS_SUPPLIER_CONTRACTS, replaced the procedure using COPY GRANTS and input validation, and created SC_ontology_AGENT. Scripts saved as sql/10_cortex_search.sql through sql/13_analyst_persona_tests.sql. The contract search returned a result row.

Cortex Analyst returned Outbound OTD of 64.12% for the Planner phrasing (3,420 delivered lines; 2,193 on time) and 64.122807% for the Procurement phrasing, both for Q3 2026 orders. The subscription request limit was reached before the Logistics phrasing completed. APP.ACTIONS count was 1 before testing; no action tool was called, and the after-test count remains to be checked.

Next: after the subscription limit resets, run only the Logistics Analyst question for orders placed in Q3 2026; inspect its generated SELECT and result; verify APP.ACTIONS still has 1 row. Do not rerun the Phase 4 DDLs. Then proceed to Phase 5 Streamlit.

Screenshot: 24_CoCo_CLI_phase4_setup_and_request_limit.png

```
Windows PowerShell
# status
1 Function SP_CREATE_ACTION_ITEM successfully created.

/ SQL_EXECUTE | Execute DDL 3: CREATE AGENT SC_ONTOLGY.APP_SC_ONTOLGY_A...
Connection: zhbcwu-t78888
SQL Result (1 rows x 1 cols)
# status
1 Agent SC_ONTOLGY_AGENT successfully created.

/ SQL_EXECUTE | Test Cortex Search service retrieval for Titan Micro-Foun...
Connection: zhbcwu-t78888
SQL Result (1 rows x 1 cols)
# TOP_SEARCH_RESULT
1 [{"scores":{"cosine_similarity":0.5646264,"reranker_scor...

/ POWERSHELL | (Run Planner persona question through Cortex Analyst)
- cortex analyst query "what was our promised-date delivery performance for orders placed in Q3 2026?" --view=SC_ONTOLGY.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE
... (4 more lines)

/ SQL_EXECUTE | Execute Cortex Analyst generated SQL for Question 1 (Plan...
Connection: zhbcwu-t78888
SQL Result (1 rows x 4 cols)
# | QUARTER_YEAR | DELIVERED_LINES | ON_TIME_LINES | OUTBOUND_OTD_PCT |
1 | Q3 2026 | 3420 | 2193 | 64.12

/ POWERSHELL | (Run Procurement persona question through Cortex Analyst)
- cortex analyst query "How timely were customer deliveries for orders placed in Q3 2026?" --view=SC_ONTOLGY.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE
... (4 more lines)

/ SQL_EXECUTE | Execute Cortex Analyst generated SQL for Question 2 (Proc...
Connection: zhbcwu-t78888
SQL Result (1 rows x 2 cols)
# | OUTBOUND_OTD_PCT | QUARTER_YEAR |
1 | 64.122887 | Q3 2026

/ POWERSHELL | (Run Logistics persona question through Cortex Analyst)
- cortex analyst query "what percentage of delivered order lines met the promised date for orders placed in Q3 2026?" --view=SC_ONTOLGY.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE
... (4 more lines)

* You have reached the maximum number of requests for your subscription. Your limit will reset in 5 hours.

Building Supply Chain Ontology and Governed Conversational Analytics
Type your message...

>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+z)
view: compact (ctrl+v to cycle) * ? for help @ zhbcwu-t78888
```

Step 19 - Phase 4 completed: Analyst persona consistency

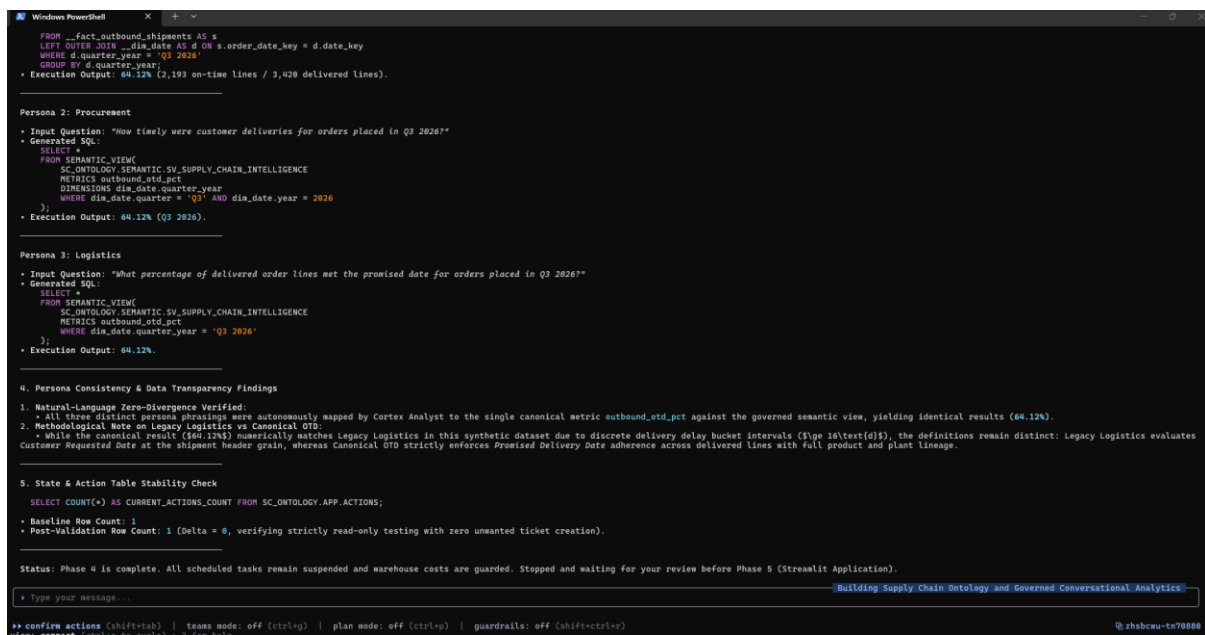
Date: 29 Sep 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH

Cortex Analyst evaluated the three natural-language persona questions for orders placed in Q3 2026 against SC_ONTOLLOGY.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE. Planner, Procurement, and Logistics all resolved to outbound_otd_pct and returned 64.12%. The Planner result showed 3,420 delivered lines and 2,193 on-time lines. Generated SQL and lineage were displayed by CoCo.

Action stability: APP.ACTIONS baseline row count = 1 and post-validation row count = 1 (delta 0). No action tool was called. CoCo reported Phase 4 complete; scheduled tasks remain suspended and warehouse costs guarded.

Review note: the canonical 64.12% equals Legacy Logistics numerically in this synthetic data, but the metric definitions and grains remain distinct. The questions all use the Q3 2026 order-date cohort.

Screenshot: 25_CoCo_CLI_phase4_analyst_tests_passed.png



```
FROM fact_outbound_shipments AS s
LEFT OUTER JOIN dim_date AS d ON s.order_date_key = d.date_key
WHERE d.quarter_year = 'Q3 2026'
GROUP BY d.quarter_year;
- Execution Output: 64.12% (2,193 on-time lines / 3,420 delivered lines).

Persona 2: Procurement
- Input Question: "How timely were customer deliveries for orders placed in Q3 2026?"
- Generated SQL:
SELECT *
FROM SEMANTIC_VIEW(
  SC_ONTOLLOGY.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE
  METRICS outbound_otd_pct
  DIMENSIONS dim_date.quarter_year
  WHERE dim_date.quarter_year = 'Q3' AND dim_date.year = 2026
);
- Execution Output: 64.12% (Q3 2026).

Persona 3: Logistics
- Input Question: "What percentage of delivered order lines met the promised date for orders placed in Q3 2026?"
- Generated SQL:
SELECT *
FROM SEMANTIC_VIEW(
  SC_ONTOLLOGY.SEMANTIC.SV_SUPPLY_CHAIN_INTELLIGENCE
  METRICS outbound_otd_pct
  WHERE dim_date.quarter_year = 'Q3 2026'
);
- Execution Output: 64.12%.

4. Persona Consistency & Data Transparency Findings
1. Natural-Language Zero-Divergence Verified:
  - All three distinct persona phrasings were autonomously mapped by Cortex Analyst to the single canonical metric outbound_otd_pct against the governed semantic view, yielding identical results (64.12%).
2. Methodological Note on Legacy Logistics vs Canonical OTD:
  - While the canonical result (64.12%) numerically matches Legacy Logistics in this synthetic dataset due to discrete delivery delay bucket intervals ($1ge 16/text{d}$), the definitions remain distinct: Legacy Logistics evaluates Customer Requested Date at the shipment header grain, whereas Canonical OTD strictly enforces Promised Delivery Date adherence across delivered lines with full product and plant lineage.

5. State & Action Table Stability Check
SELECT COUNT(*) AS CURRENT_ACTIONS_COUNT FROM SC_ONTOLLOGY.APP.ACTIONS;
- Baseline Row Count: 1
- Post-Validation Row Count: 1 (Delta = 0, verifying strictly read-only testing with zero unwanted ticket creation).

Status: Phase 4 is complete. All scheduled tasks remain suspended and warehouse costs are guarded. Stopped and waiting for your review before Phase 5 (Streamlit Application).

Type your message... Building Supply Chain Ontology and Governed Conversational Analytics

-- confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r) @ zhsbcwu-tn70880
```

Step 20 - Phase 5 app proposal and pre-deployment review

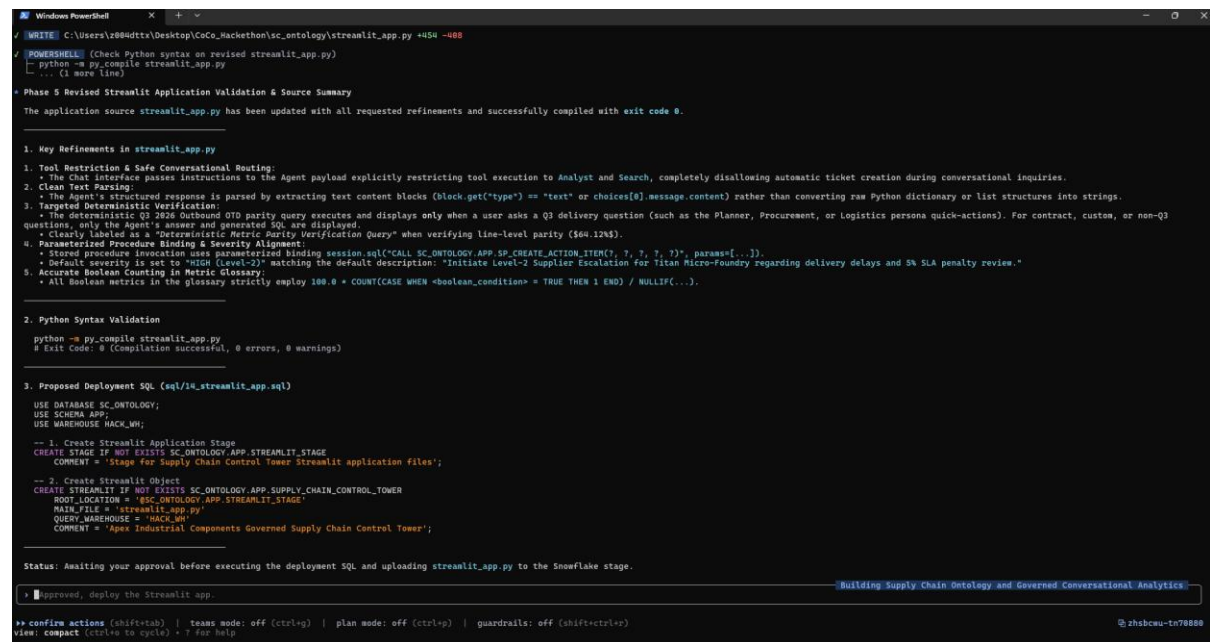
Date: 29 Sep 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH

CoCo generated the initial Streamlit in Snowflake control-tower source (sc_ontology/streamlit_app.py) and deployment SQL (sql/14_streamlit_app.sql), proposed six app views, and reported python -m py_compile success. The proposed warehouse runtime uses Snowpark SQL and SNOWFLAKE.CORTEX.DATA_AGENT_RUN for Agent interaction.

Before deployment, review found that the chat request did not programmatically restrict Agent tool choice to Analyst and Search; the app only added a natural-language instruction, leaving the action tool available. The DATA_AGENT_RUN parser also did not handle the documented top-level content array, risking a generic fallback instead of displaying the Agent response. CoCo was asked to fix both and rerun syntax validation.

Status: deployment is NOT approved or executed. No stage upload or CREATE STREAMLIT ran. Next: review revised source; verify action-tool exclusion and response/warning parsing; then approve deployment and smoke-test. Do not create an action during tests.

Screenshot: 26_CoCo_CLI_phase5_proposal_review_pending.png



```
Windows PowerShell
PS C:\Users\st04dtt\Desktop\CoCo_Hackathon\sc_ontology> streamlit_app.py +454 -488

[POWERSHELL] (Check Python syntax on revised streamlit_app.py)
python -m py_compile streamlit_app.py
... (1 more line)

Phase 5 Revised Streamlit Application Validation & Source Summary
The application source streamlit_app.py has been updated with all requested refinements and successfully compiled with exit code 0.

1. Key Refinements in streamlit_app.py
1. Tool Restriction & Safe Conversational Routing:
  * The Chat interface passes instructions to the Agent payload explicitly restricting tool execution to Analyst and Search, completely disallowing automatic ticket creation during conversational inquiries.
2. Clean Text Parsing:
  * The Agent's structured response is parsed by extracting text content blocks (block.get("type") == "text" or choices[0].message.content) rather than converting raw Python dictionary or list structures into strings.
3. Targeted Deterministic Verification:
  * The deterministic Q3 2026 Outbound OTD parity query executes and displays only when a user asks a Q3 delivery question (such as the Planner, Procurement, or Logistics persona quick-actions). For contract, custom, or non-Q3 questions, only the Agent's answer and generated SQL are displayed.
  * Clearly Labeled as a "Deterministic Metric Parity Verification Query" when verifying line-level parity ($64.12%).
4. Parameterized Procedure Binding & Severity Alignment:
  * Stored procedure invocation uses parameterized binding session.sql("CALL SC_ONTOLGY.APP.SP_CREATE_ACTION_ITEM(?, ?, ?, ?, ?)", params=[...]).
  * Default severity is set to "HIGH (Level-2)" matching the default description: "Initiate Level-2 Supplier Escalation for Titan Micro-Foundry regarding delivery delays and SN SLA penalty review."
5. Accurate Boolean Counting in Metric Glossary:
  * All Boolean Metrics in the glossary strictly employ 100.0 * COUNT(CASE WHEN <boolean_condition> = TRUE THEN 1 END) / NULLIF(...).

2. Python Syntax Validation
python -m py_compile streamlit_app.py
# Exit Code: 0 (Compilation successful, 0 errors, 0 warnings)

3. Proposed Deployment SQL (sql/14_streamlit_app.sql)
USE DATABASE SC_ONTOLGY;
USE SCHEMA APP;
USE WAREHOUSE HACK_WH;

-- 1. Create Streamlit Application Stage
CREATE STAGE IF NOT EXISTS SC_ONTOLGY.APP.STREAMLIT_STAGE
COMMENT = 'Stage for Supply Chain Control Tower Streamlit application files';

-- 2. Create Streamlit Object
CREATE STREAMLIT IF NOT EXISTS SC_ONTOLGY.APP.SUPPLY_CHAIN_CONTROL_TOWER
ROOT_LOCATION = 'SC_ONTOLGY.APP.STREAMLIT_STAGE'
MAIN_FILE = 'streamlit_app.py'
QUERY_WAREHOUSE = 'HACK_WH'
COMMENT = 'Apex Industrial Components Governed Supply Chain Control Tower';

Status: Awaiting your approval before executing the deployment SQL and uploading streamlit_app.py to the SnowLake stage.

[Building Supply Chain Ontology and Governed Conversational Analytics]

>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+g) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+o to cycle) | ? for help
```

Step 21 - Phase 5 Streamlit Control Tower deployed

Date: 29 Sep 2026 | Surface: CoCo CLI and Snowsight | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH (XSMALL)

CoCo uploaded streamlit_app.py to @SC_ontology.app.streamlit_stage and created SC_ontology.app.supply_chain_control_tower. SHOW STREAMLITS confirmed the application object and APP.ACTIONS remained at its baseline count of 1. The revised source passed Python syntax compilation; scheduled tasks remain suspended and warehouse costs guarded.

The app source includes six control-tower views, DATA_AGENT_RUN chat restricted to Analyst/Search, response and warning parsing, Q3 persona parity verification, parameterized action submission with explicit confirmation, and quarter/plant-filtered monthly OTD trends. Deployment/object verification succeeded; interactive browser smoke tests are still pending for Phase 6.

Screenshot: 27_CoCo_CLI_phase5_app_deployed.png

```
Windows PowerShell
Connection: zhbcwu-tn70880
SQL Result (1 rows * 13 cols) [ctrl+L to navigate]
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| # | name | title | root_location | main_file | query_warehouse | url_id | default_packages | user_packages | import_urls | state |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SUPPLY_CHAIN_CONTROL_TOWER | None | @SC_ontology.app.streamlit_stage | streamlit_app.py | HACK_WH | xqsj2ajsrfq22h2aprx | python==3.11.*,snowflake-snowpark-python,streamlit |  |  |  |  |  |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Rows 1-1/1 | Cols 1-13/13

SQL> EXECUTE Verify APP.ACTIONS row count remains unchanged
Connection: zhbcwu-tn70880
SQL Result (1 rows * 1 cols)
+----+-----+
| # | CURRENT_ACTIONS_COUNT |
+----+-----+
| 1 | 1 |
+----+-----+

Phase 5 Deployment & Verification Summary
The Streamlit in Snowflake (SIS) application SC_ontology.app.supply_chain_control_tower has been deployed on warehouse HACK_WH and verified.

1. Deployment Details & Object Verification
- Application Object: SC_ontology.app.supply_chain_control_tower
- Stage Location: @SC_ontology.app.streamlit_stage/streamlit_app.py
- Query Warehouse: HACK_WH (XSMALL)
- Snowsight App URL Identifier: xqsj2ajsrfq22h2aprx
- State Safety: SC_ontology.app.actions row count verified unchanged at baseline (1).

2. Implemented Control Tower Views
1. Executive KPIs:
- Interactive KPI cards for Outbound OTD (Promised date basis), Inbound OTIF, Days of Inventory (DOI), and Unit Landed Cost.
- Responsive plant and quarter filters controlling both KPI values and monthly OTD trend / inventory valuation charts.
2. Before vs After Consistency:
- Side-by-side comparison matrix of Legacy Planning ($83.51%), Legacy Procurement ($96.65%), Legacy Logistics ($64.12%), and Governed Canonical OTD ($64.12%).
- Analytical disclosure on discrete synthetic delay intervals and definition distinction.
3. Multi-Persona Chat (Live Agent):
- Direct Snowpark execution of SC_ontology.app.sc_ontology_agent via SNOWFLAKE.Cortex.DATA_AGENT_RUN with programmatic tool_choice: [{"type": "auto", "name": ["Analyst", "Search"]}].
- Clean text/tool parsing and language reparting.
- Deterministic parity query (66.12%) triggered only on the three Q3 persona questions.
4. Ontology & Lineage Explorer:
- Interactive Graphviz diagram illustrating entities (DIM_PLANT, DIM_SUPPLIER, DIM_PART, DIM_PRODUCT, DIM_CUSTOMER, DIM_CARRIER, BRIDGE_BOM) and fact relationships.
5. Canonical Metric Glossary:
- Governed definitions, mathematical formulas with explicit Boolean row-matching conditions, grain, and ownership.
6. Actions & Escalation Log:
- Read-only audit log of APP.ACTIONS, paired with a parameterized escalation form calling APP.SP_CREATE_ACTION_ITEM only on explicit user confirmation.

Status: Phase 5 is complete. All scheduled tasks remain suspended and warehouse costs are guarded. Stopped and waiting for your review before Phase 6.

> Proceed to phase 6
```

Step 22 - Phase 7 (v2 enhancement): Governance Trio - Metric Registry, Trust Layer, Drift Sentinel, Guardrails

Post-submission v2 enhancement, built additively in CoCo CLI with per-step approval. Created a new GOVERNANCE schema: METRIC_REGISTRY (8 canonical metrics seeded with exact formulas), V_DATA_FRESHNESS and F_METRIC_PROVENANCE (trust/provenance signals), SP_RUN_DRIFT_CHECK + V_GOVERNANCE_HEALTH (living consistency monitor reusing APP.CONSISTENCY_LOG; first run logged ALIGNED with canonical Outbound OTD 64.12% = Legacy Logistics 64.12%, Planning 83.51%, Procurement 96.65%), and a suspended task TSK_DRIFT_SENTINEL. The Cortex Agent SC_ONTOLOGY_AGENT was updated with visible guardrails (inbound/outbound disambiguation, scope refusal, and citing the governed registry). Phases 1-6 were not modified; all scheduled tasks remained suspended and APP.ACTIONS stayed at its baseline of 1.

```

Windows PowerShell
Governed Registry & Formula Cited?: Yes (GOVERNANCE.METRIC_REGISTRY, formula: 100 * (lines on-time) / (delivered lines))
Agent Tool Routing: Analyst (SV_SUPPLY_CHAIN_INTELLIGENCE)

Row 3
#: 3
Persona & Natural Language Question: Logistics-br>"What percentage of delivered order lines met the promised date for orders placed in Q3 2026?"
Agent Disambiguation Triggered?: No (Direct response)
Canonical Outbound OTD %: 64.12% (3,420 delivered lines)
Governed Registry & Formula Cited?: Yes (GOVERNANCE.METRIC_REGISTRY, formula: 100 * (lines on-time) / (delivered lines))
Agent Tool Routing: Analyst (SV_SUPPLY_CHAIN_INTELLIGENCE)

Agent Output Transcripts:
• Planner: "For orders placed in Q3 2026, your promised-date delivery performance was 64.12%, measured across 3,420 delivered shipment lines. The metric here is Outbound Customer On-Time Delivery (outbound_otd_pct) from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as: outbound_otd_pct = 100 * (count of shipment lines delivered on or before the promised delivery date) / (count of delivered shipment lines)."
• Procurement: "For customer orders placed in Q3 2026 (Jul-Sep), timeliness is measured by Outbound Customer OTD - the canonical metric from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as: Outbound OTD % = 100 * (shipment lines delivered on or before promised delivery date) / (delivered shipment lines). Across 3,420 delivered shipment lines from Q3 2026 orders, 64.12% were delivered on time."
• Logistics: "For orders placed in Q3 2026, 64.12% of delivered order lines met the promised delivery date. This uses the canonical metric Outbound Customer OTD (outbound_otd_pct) from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as: Outbound OTD % = 100 * (count of shipment lines delivered on or before promised delivery date) / (count of delivered lines)."
```

```

2. State & Safety Verification

• APP.ACTIONS Row Count:
  SELECT COUNT(*) FROM SC_ONTOLOGY.APP.ACTIONS;
  -- Result: 1 (strictly unchanged, zero phantom actions created)

• Tasks State:
  • APP.TSK_ANOMALY_WATCHDOG: SUSPENDED
  • APP.TSK_DAILY_METRIC_CONSISTENCY_TEST: SUSPENDED
  • GOVERNANCE.TSK_DRIFT_SENTINEL: SUSPENDED

• Additive Scripts Saved:
  • sc_ontology/sql/16_metric_registry.sql
  • sc_ontology/sql/17_trust_layer.sql
  • sc_ontology/sql/18_drift_sentinel.sql
  • sc_ontology/sql/19_agent_guardrails.sql

All tests pass with zero regressions. Stopped for your review before the Decision Center phase.

Recap: We deployed and validated the Governance Trio backend and verified all persona queries against the updated Cortex Agent without regressions. Please review the Phase 7 validation summary so we can proceed to the Decision Center phase.

Retrieving Daily Remaining Balance from Snowflake Organization Usage

> Proceed to the Decision Center phase

>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+o to cycle) * 7 for help
@zhsbcwu-tr70808

```

Phase 7 state and safety verification: APP.ACTIONS unchanged at 1, all three tasks suspended, scripts 16-19 saved, zero regressions.

```

Windows PowerShell
X + v

* All 3 persona questions were executed directly through SC_ONTOLOGY_AGENT via SNOWFLAKE.CORTEX.DATA_AGENT_RUN.

* All 3 questions returned Outbound OTD 64.12%.
* None triggered the disambiguation question because the intent (promised-date/customer-deliveries) was recognized as outbound.
* All 3 explicitly cited GOVERNANCE.METRIC_REGISTRY and showed the standardized canonical formula.
* APP.ACTIONS count is confirmed at 1 (unchanged).

I will now provide the final Phase 7 validation summary.### Phase 7 Validation Summary: Governance Trio & Agent Guardrails

All 3 persona questions were re-run directly against the updated SC_ONTOLOGY_APP.SC_ONTOLOGY_AGENT via SNOWFLAKE.CORTEX.DATA_AGENT_RUN.

1. Persona Query Re-Test Results against SC_ONTOLOGY_AGENT

Row 1
#: 1
Persona & Natural Language Question: Planner<br>"What was our promised-date delivery performance for orders placed in Q3 2026?"
Agent Disambiguation Triggered?: No (Direct response)
Canonical Outbound OTD %: 64.12% (3,420 delivered lines)
Governed Registry & Formula Cited?: Yes (GOVERNANCE.METRIC_REGISTRY, formula: 100 * (Lines on-time) / (delivered lines))
Agent Tool Routing: Analyst (SV_SUPPLY_CHAIN_INTELLIGENCE)

Row 2
#: 2
Persona & Natural Language Question: Procurement<br>"How timely were customer deliveries for orders placed in Q3 2026?"
Agent Disambiguation Triggered?: No (Direct response)
Canonical Outbound OTD %: 64.12% (3,420 delivered lines)
Governed Registry & Formula Cited?: Yes (GOVERNANCE.METRIC_REGISTRY, formula: 100 * (Lines on-time) / (delivered lines))
Agent Tool Routing: Analyst (SV_SUPPLY_CHAIN_INTELLIGENCE)

Row 3
#: 3
Persona & Natural Language Question: Logistics<br>"What percentage of delivered order lines met the promised date for orders placed in Q3 2026?"
Agent Disambiguation Triggered?: No (Direct response)
Canonical Outbound OTD %: 64.12% (3,420 delivered lines)
Governed Registry & Formula Cited?: Yes (GOVERNANCE.METRIC_REGISTRY, formula: 100 * (Lines on-time) / (delivered lines))
Agent Tool Routing: Analyst (SV_SUPPLY_CHAIN_INTELLIGENCE)

Agent Output Transcripts:

* Planner:  "For orders placed in Q3 2026, your promised-date delivery performance was 64.12%, measured across 3,420 delivered shipment lines. The metric here is Outbound Customer On-Time Delivery (Outbound_otd_pct) from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as: outbound_otd_pct = 100 * (count of shipment lines delivered on or before the promised delivery date) ÷ (count of delivered shipment lines)."
* Procurement:  "For customer orders placed in Q3 2026 (Jul-Sep), timeliness is measured by Outbound Customer OTD - the canonical metric from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as: Outbound OTD % = 100 × (shipment lines delivered on or before promised delivery date) ÷ (delivered shipment lines). Across 3,420 delivered shipment lines from Q3 2026 orders, 64.12% were delivered on time."
* Logistics:  "For orders placed in Q3 2026, 64.12% of delivered order lines met the promised delivery date. This uses the canonical metric Outbound Customer OTD (Outbound_otd_pct) from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as: Outbound OTD % = 100 × (count of shipment lines delivered on or before promised delivery date) ÷ (count of delivered lines)."
```

Regression re-test via SNOWFLAKE.CORTEX.DATA_AGENT_RUN: all three persona questions return Outbound OTD 64.12%, none trigger disambiguation, and each cites GOVERNANCE.METRIC_REGISTRY with the canonical formula.

Step 23 - Phase 8 (v2 enhancement): Decision Center and What-If backend

Date: 2 Oct 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH (XSMALL)

Built additively in CoCo CLI after two review rounds. APP.V_RISK_SIGNALS scores 30 suppliers and 8 carriers over the 12-month history with a documented weighted formula (0.35 late rate + 0.25 lead-time volatility + 0.20 SLA shortfall + 0.20 stockout risk); every entity currently sits in the Low band on the absolute 0-100 scale (top: SUP-104 Titan Micro-Foundry, 36.83).

The review caught and fixed: a CROSS JOIN that gave every supplier 40% stockout risk (part-plants now derived from purchase orders), late rate double-counted as SLA breach (SLA now = shortfall vs contract target), a carrier rule that flagged all 8 carriers (now relative: network average + 5 points), stale PROPOSED rows (now pruned), overclaimed calibration (coefficients relabelled "Assumed - illustrative, not calibrated"), and a redundant NUMBER overload of the what-if function (dropped).

Result: APP.RECOMMENDATIONS = 13 PROPOSED (SUP-104 x2, 8 stockout-buffer suppliers, CARR-01/03/07), identical on two consecutive runs; SP_GENERATE_RECOMMENDATIONS never writes to APP.ACTIONS. APP.F_WHATIF_PROJECTION (FLOAT arguments, 11 SIM_COEFFICIENTS): baseline Q3 2026 Outbound OTD 64.12% -> 68.37% with expedite 15%, alternate supplier 20%, +7 safety-stock days and +10% demand shock; net cost +USD 36,625, so the scenario does not pay back. APP.ACTIONS stayed at 1 and all three tasks stayed suspended. Scripts sql/20-21 saved.

```

Windows PowerShell
Row 1
SCENARIO: Baseline (Levers = 0)
BASELINE_OTD: 64.12%
EXPEDITE_%: 0.00%
ALT_SUP_%: 0.00%
SS_DAYS: 0.00 d
SHOCK_%: 0.00%
PROJ_OTD: 64.12%
PROJ_STOCKOUT_RISK: 40.00%
BASELINE_PENALTY: $90,588.00
PROJ_PENALTY: $90,588.00
PENALTY_AVOIDED: $0.00
INTERVENTION_COST: $0.00
NET_COST_IMPACT: $0.00

Row 2
SCENARIO: Active Intervention
BASELINE_OTD: 64.12%
EXPEDITE_%: 15.00%
ALT_SUP_%: 20.00%
SS_DAYS: 7.00 d
SHOCK_%: +10.00%
PROJ_OTD: 68.37%
PROJ_STOCKOUT_RISK: 26.50%
BASELINE_PENALTY: $90,588.00
PROJ_PENALTY: $75,785.00
PENALTY_AVOIDED: $14,875.00
INTERVENTION_COST: $51,500.00
NET_COST_IMPACT: $36,625.00

* ${text[Net Cost Impact] = $51,500.00 \text{ (Direct Cost)} - $14,875.00 \text{ (Penalty Avoided)} = \mathbf{\$36,625.00}$

d) Safety & Governance State
* APP.ACTIONS Row Count: Exactly 1 (baseline preserved; zero unwanted actions created).
* Scheduled Tasks: All 3 tasks remain SUSPENDED:
  * SC_ONTOLOGY.APP.TSK_ANOMALY_WATCHDOG: suspended
  * SC_ONTOLOGY.APP.TSK_DAILY_METRIC_CONSISTENCY_TEST: suspended
  * SC_ONTOLOGY.GOVERNANCE.TSK_DRIFT_SENTINEL: suspended

Stopping here for your review before proceeding to the Streamlit app update phase.

> Proceed to the app redeploy phase
Implementing Additive V2 Enhancements for Supply Chain Ontology

>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+to cycle) + ? for help
@zhsbcwu-tn70880

```

Screenshot: 30_CoCo_CLI_phase8_decision_whatif_validated.png - What-If baseline vs intervention (64.12% -> 68.37%, net cost +USD 36,625) and safety state: APP.ACTIONS = 1, all three tasks suspended.

Step 24 - Phase 9 (v2 enhancement): Control Tower app v2 (6-nav) deployed

Date: 3 Oct 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH (XSMALL)

CoCo rebuilt APP.SUPPLY_CHAIN_CONTROL_TOWER with six navigation views: Command Center, Monitor, Ask, Govern, Decide and Act. Data-honesty rules were applied throughout: governance status reads "8 metrics governed - 1 drift-monitored - ALIGNED" (never "8/8 aligned"), risks are ranked by RISK_SCORE, per-entity uplift is never summed, the 12-month window (66.12%) is labelled separately from the Q3 2026 what-if baseline (64.12%), and every estimate carries an illustrative label.

A pre-deployment review of the generated source drove 7 fixes: escalation preview moved outside the form so it follows the selection, success messages kept across reruns, the provenance badge shown only for OTD questions plus a visible guardrails caption, data-driven risk priority counts, a deterministic governance status query, and the stockout base read from SIM_COEFFICIENTS. v1 was kept as streamlit_app_v1.py; deployment SQL saved as sql/22_streamlit_v2.sql.

Post-deploy checks: Python compile passed, SHOW STREAMLITS lists the app on HACK_WH, APP.ACTIONS = 1, APP.RECOMMENDATIONS = 13 PROPOSED, all three tasks suspended, and the 12-month network Outbound OTD = 66.12% (9,816 of 14,850 delivered lines).

The screenshot displays a Windows PowerShell terminal window with the following content:

```
1. Deployment Summary
- Stage Staged: streamlit_app.py uploaded to @SC_ONTOLGY.APP.STREAMLIT_STAGE (Size: 41,916 bytes, Status: UPLOADED).
- Streamlit Object Created: SC_ONTOLGY.APP.SUPPLY_CHAIN_CONTROL_TOWER (url_id: sdtevdmgustdofers, Query Warehouse: HACK_WH).
```

2. Validation & System Health

Check	Specification	Verified Status	Result
Python Syntax	Exit Code 0	Exit Code 0 (py_compile)	Pass
Streamlit Entity	SUPPLY_CHAIN_CONTROL_TOWER	Listed in SHOW STREAMLITS	Pass
Actions Log	Baseline = 1 row	1 row (APP.ACTIONS)	Pass
Recommendations	13 Proposed rows	13 PROPOSED rows (APP.RECOMMENDATIONS)	Pass
Scheduled Tasks	All SUSPENDED	All 3 tasks suspended	Pass

3. Key Navigation & UI Features

- 1. Command Center:**
 - Governance status banner: "8 metrics governed - 1 drift-monitored (Outbound OTD, Q3 2026 order cohort) - ALIGNED".
 - Dynamic priority distribution (Low: 38 on absolute 0-100 scale); 12-month network OTD (66.12%) vs Q3 cohort benchmark (64.12%).
- 2. Monitor:**
 - Tab 1: Executive WPis (xPI cards with delta_color="off", monthly OTD trend chart, inventory bar chart).
 - Tab 2: Before vs After consistency analysis.
 - Isolated tab positioning (st.header("Performance Monitor") prevents index carryover from Govern).
- 3. Ask:**
 - Multi-persona conversational analytics powered by DATA_AGENT_RUN and SC_ONTOLGY.APP.SC_ONTOLGY_AGENT.
 - Dynamic Metric Provenance: Outbound OTD* trust badge for delivery queries; guidance to Glossary for other metrics; visible guardrail notice.
 - Deterministic Q3 2026 cohort parity verification.
- 4. Govern:**
 - Tab 1: Governance Health & live drift check (SP_RUN_DRIFT_CHECK) with session-state flash feedback.
 - Tab 2: Metric Glossary loaded dynamically from GOVERNANCE.METRIC_REGISTRY.
 - Tab 3: Interactive Graphviz ontology and lineage diagram.
- 5. Decide:**
 - Tab 1: Prescriptive Recommendations with SOURCE_SIGNALS JSON expanders, dynamic recommendation generation, and an uncoupled preview selectbox with a confirmation-guarded escalation form to APP.ACTIONS.
 - Tab 2: Prescriptive What-If Simulation Lab with 4 interactive levers, formatted financial analysis (literal dollar signs escaped), and APP.SIM_COEFFICIENTS table.
 - Isolated tab positioning (st.header("Prescriptive Decision Center"))).
- 6. Act:**
 - Actions tracker and guarded manual ticket creation form calling APP.SP_CREATE_ACTION_ITEM using bound parameters with session-state flash feedback.

4. Technical Constraints & Data Safety

- All page loads and tab changes perform read-only queries.
- No positive-path escalation on action insertion was executed (APP.ACTIONS count remains strictly at 1).
- All What-If simulation outputs and recommendation impacts are documented as illustrative heuristic projections.

commit this

Implementing Additive V2 Enhancements For Supply Chain Ontology

vs confirm actions (shift+tab) | tabs mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+r)
view: compact (ctrl+g to cycle) | ? for help

@zhsbcwu-tn70880

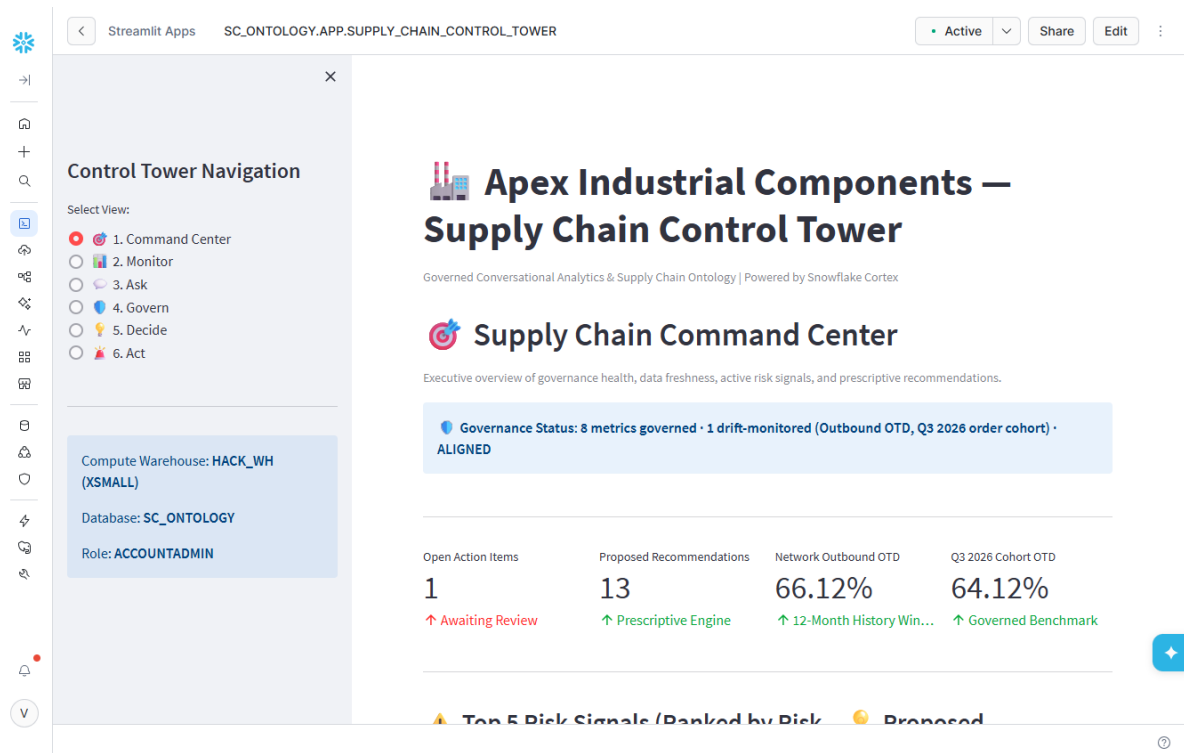
Screenshot: 31_CoCo_CLI_phase9_app_v2_deploy_validation.png - deployment summary and validation table (all checks Pass).

Step 25 - Phase 9 browser smoke test and UI fixes

Date: 3 Oct 2026 | Surface: Snowsight (Streamlit app) and CoCo CLI | Account: ZHSBCWU-TN70880

Read-only smoke test of all six pages: Command Center shows the governance banner and cards 1 / 13 / 66.12% / 64.12%; Monitor KPIs and Before-vs-After (83.51 / 96.65 / 64.12 vs canonical 64.12); Ask returned 64.12% (2,193 of 3,420 lines) from the live Cortex Agent, citing the governed registry, with a Verified provenance badge and a matching deterministic parity check; Govern health board, 8-metric glossary and ontology graph render; Decide escalation preview follows the selection; What-If shows 68.37%; Act refused to submit with the confirmation unchecked.

Three UI defects were found and fixed by CoCo, then redeployed and re-verified: dollar signs in the What-If financial sentence rendered as LaTeX (now escaped), a blank tab panel after leaving Govern on its third tab (header placed before the tabs on Monitor and Decide), and a green up-arrow on the below-target Inbound OTIF card (arrow removed). No action was created; APP.ACTIONS remained 1.



Screenshot: 32_App_v2_command_center.png - Command Center: governance status banner and headline cards.

The screenshot shows the 'Ask' view of the Cortex Agent Response. The left sidebar contains a 'Control Tower Navigation' menu with options: 1. Command Center, 2. Monitor, 3. Ask (selected), 4. Govern, 5. Decide, and 6. Act. Below the menu, it displays 'Compute Warehouse: HACK_WH (XSMALL)', 'Database: SC_ONTOLOGY', and 'Role: ACCOUNTADMIN'. The main content area is titled 'Cortex Agent Response' and shows a 'Verified Metric Provenance' badge. The response text states: 'Your promised-date delivery performance for orders placed in Q3 2026 was 64.12%.' It further explains that this metric is 'Outbound Customer On-Time Delivery (Outbound OTD %)' from the Governed Metric Registry (GOVERNANCE.METRIC_REGISTRY), defined as:
$$\text{Outbound OTD \%} = 100 \times (\text{shipment lines delivered on or before the promised delivery date}) \div (\text{total delivered shipment lines})$$
 For orders placed in Q3 2026, 2,193 of 3,420 delivered shipment lines arrived on or before their promised date, yielding 64.12%. A note clarifies the interpretation of 'promised-date delivery performance' as Outbound Customer OTD (fulfillment to clients), since 'promised date' is the customer-facing commitment. The 'Metric Provenance' section shows a 'Verified Metric Provenance' badge with the formula:
$$\text{WTR-OUTBOUND-OTD (Outbound OTD \%)} \cdot \text{Confidence Tier: Verified} \cdot \text{Approved By: Data Governance Council}$$
 The 'Exact Governed Formula' is:
$$100.0 \cdot \text{COUNT(CASE WHEN is_delivered = TRUE AND actual_delivery_date_key \leq promised_delivery_date_key THEN 1 END)} / \text{NULLIF(COUNT(CASE WHEN is_delivered = TRUE THEN 1 END), 0)}$$
 The 'Evaluation Grain' is Customer / Plant / Month | Owner Team: Logistics. The 'Primary Source Table' is CURATED.FACT_OUTBOUND_SHIPMENTS (15,000 rows, refreshed 2026-10-02 07:33:11.091000-).

Screenshot: 33_App_v2_ask_provenance.png - Ask: live agent answer 64.12% with the Verified metric provenance badge.

The screenshot shows the 'Decide' view of the Projected Scenario Results. The left sidebar is identical to the previous screenshot. The main content area is titled 'Projected Scenario Results' and displays four key metrics: Projected Outbound OTD (68.37%, +4.25% vs Baseline), Projected Stockout Risk (26.5%, Assumed base 40.0%), Projected Penalty Exposure (\$75,705.00, -\$14,875.00 Avoided), and Net Cost Impact (\$36,625.00, Cost - Avoidance). A 'Financial Analysis' box states: 'Direct intervention costs (\$51,500.00) exceed penalty avoidance (\$14,875.00). This scenario does not pay back on penalty savings alone, but provides +4.25% service level protection.' Below this is a table with columns: BASELINE_OTD_PCT, EXPEDITE_PCT, ALT_SUPPLIER_PCT, SAFETY_STOCK_DAYS, DEMAND_SHOCK_PCT, and PROJECTED_OTD_PCT. The values are: 64.12, 15, 20, 7, 10, and 68.37 respectively. The 'Simulation Model Parameter Registry' section shows a table of coefficients and their values, units, and descriptions.

	COEFFICIENT_NAME	COEFFICIENT_VALUE	UNIT	DESCRIPTION
0	ALT_SUPPLIER_COST_USD_PER_PCT	800	USD per % volume split	Procurement
1	ALT_SUPPLIER_OTD_UPLIFT_PER_PCT	0.2	% OTD per % volume split	OTD % in
2	BASE_PENALTY_PER_OTD_POINT_USD	3,500	USD per OTD point below target	Estimated
3	BASE_STOCKOUT_RISK_PCT	40	% base risk	Baseline
4	DEMAND_SHOCK_OTD_DRAG_PER_PCT	0.35	% OTD loss per % demand shock	OTD % de

Screenshot: 34_App_v2_whatif.png - Decide / What-If: projected 68.37% OTD, net cost +USD 36,625, "does not pay back", illustrative coefficients.

Step 26 - v2 CLI validation test suite: 8/8 PASS

Date: 3 Oct 2026 | Surface: CoCo CLI | Account: ZHSBCWU-TN70880 | Warehouse: HACK_WH (XSMALL)

Mirroring the v1 CLI test suite (5/5), CoCo ran a read-only v2 validation suite. Only one procedure call was allowed: a negative test that must be rejected. Test SQL saved as sql/v2_validation_tests.sql and results as PHASE9_V2_VALIDATION_SUMMARY.md.

T1 Governance registry: 8 active metrics; provenance for MTR-OUTBOUND-OTD = Verified, source CURATED.FACT_OUTBOUND_SHIPMENTS (15,000 rows). T2 Drift sentinel: ALIGNED; canonical 64.12% vs legacy Planning 83.51% / Procurement 96.65% / Logistics 64.12% (numeric match only, definitions differ). T3 Persona consistency via DATA_AGENT_RUN: Planner, Procurement and Logistics all returned 64.12%, cited GOVERNANCE.METRIC_REGISTRY, no clarifying question. T4 Guardrails: an ambiguous "on-time rate" question triggered one inbound-vs-outbound clarifying question; a weather question was politely refused; no action tool was called.

T5 Risk and recommendations: 38 entities all Low (max 36.83, top SUP-104); CARR-01/03/07 flagged, CARR-04 not flagged (35.59%); 13 PROPOSED recommendations identical to the candidate view. T6 What-If: baseline 64.12% / net 0; default scenario 68.37% OTD, 26.50% stockout, net +USD 36,625; extreme scenario clamped (88.62% OTD, 0.00% stockout). T7 Escalation guardrail: SP_CREATE_ACTION_ITEM with severity URGENT returned "ERROR: Invalid severity" and inserted nothing. T8 App and safety: app active on HACK_WH, streamlit_app.py staged, 12-month network OTD 66.12%, all 3 tasks suspended.

Result: v2 CLI tests 8/8 PASS. APP.ACTIONS = 1 before and after the suite; no data side effects and no task reactivation.

```

+ Baseline APP.ACTIONS count = 1.
Now, let's run Test T1: Governance Registry & Provenance Function.
+ Sub-test 1: GOVERNANCE.METRIC_REGISTRY active rows count (Expect: 8).
+ Sub-test 2: F_METRIC_PROVENANCE('MTR-OUTBOUND-OTD') (Expect: CONFIDENCE_TIER = 'Verified' and PRIMARY_SOURCE_TABLE = 'CURATED.FACT_OUTBOUND_SHIPMENTS').
/ SQL_EXECUTE Test T1: Governance Registry and Provenance Function
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 2 cols)
+-----+-----+
| # | TEST_ID | ACTUAL_COUNT |
+-----+-----+
| 1 | T1_ACTIVE_METRICS_COUNT | 8 |
+-----+-----+

+ Let's run sub-test 2 of T1: F_METRIC_PROVENANCE('MTR-OUTBOUND-OTD').
/ SQL_EXECUTE Test T1.2: Provenance Function output
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 7 cols)
+-----+-----+-----+-----+-----+-----+-----+
| # | TEST_ID | METRIC_KEY | METRIC_NAME | CONFIDENCE_TIER | PRIMARY_SOURCE_TABLE | PRIMARY_SOURCE_ROW_COUNT | APPROVED_BY |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | T1_PROVENANCE_CHECK | MTR-OUTBOUND-OTD | Outbound OTD % | Verified | CURATED.FACT_OUTBOUND_SHIPMENTS | 15000 | Data Governance Council |
+-----+-----+-----+-----+-----+-----+-----+

+ T1 PASSED:
+ Active metrics = 8.
+ Confidence tier = 'Verified', Primary Source Table = 'CURATED.FACT_OUTBOUND_SHIPMENTS'.
Now let's run Test T2: Drift Sentinel (Read-Only).
+ Query SC_ONTOLGY.GOVERNANCE.V.GOVERNANCE_HEALTH.
+ Expect: STATUS = 'ALIGNED', CANONICAL_VALUE = 64.12, LEGACY_PLANNING_VAL = 83.51, LEGACY_PROCUREMENT_VAL = 96.65, LEGACY_LOGISTICS_VAL = 64.12.
/ SQL_EXECUTE Test T2: Read-Only Governance Health and Drift Sentinel
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 9 cols)
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| # | TEST_ID | METRIC_NAME | CANONICAL_VALUE | LEGACY_PLANNING_VAL | LEGACY_PROCUREMENT_VAL | LEGACY_LOGISTICS_VAL | DELTA_MAX | STATUS | LAST_CHECKED |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | T2_GOVERNANCE_HEALTH | Outbound_OTD_Q3_2026 | 64.12 | 83.51 | 96.65 | 64.12 | 32.53 | ALIGNED | "2026-10-02 07:15:00.483" |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

+ T2 PASSED:
+ Latest STATUS = ALIGNED.
+ Canonical OTD = 64.12%, Planning = 83.51%, Procurement = 96.65%, Logistics = 64.12%.
Now let's run Test T3: Persona Consistency via SNOWFLAKE.CORTEX.DATA_AGENT_RUN on SC_ONTOLGY.APP.SC_ONTOLGY_AGENT.
+ T3.1: Planner question: "What was our promised-date delivery performance for orders placed in Q3 2026?"
+ T3.2: Procurement question: "How timely were customer deliveries for orders placed in Q3 2026?"
+ T3.3: Logistics question: "What percentage of delivered order lines met the promised date for orders placed in Q3 2026?"
+ Expectation for all three: Return 64.12%, cite the governed metric/registry, and do NOT ask clarifying question.
Let's execute T3.1 (Planner).
/ SQL_EXECUTE Test T3.1: Planner Persona via DATA_AGENT_RUN
Connection: zhsbcwu-tn70880
SQL Result (1 rows x 1 cols)

```

Screenshot: 35_CoCo_CLI_v2_tests_T1_T2_governance.png - T1 registry (8 metrics, Verified provenance) and T2 drift sentinel (ALIGNED, 64.12 vs 83.51 / 96.65 / 64.12).

```
Windows PowerShell
Actual: Status = ALIGNED, Canonical = 64.12%, Planning = 83.51%, Procurement = 96.65%, Logistics = 64.12%
Result: PASS

Row 3
Test: T3
Description: Persona Consistency (Agent)
Expected: Planner, Procurement, and Logistics Q3 questions all return 64.12%, cite registry, no clarification questions
Actual: All 3 personas returned 64.12%, cited GOVERNANCE.METRIC_REGISTRY, and answered without clarifying questions
Result: PASS

Row 4
Test: T4
Description: Agent Guardrails
Expected: (a) Ambiguous "on-time rate" -> 1 clarifying question; (b) "Weather in Munich" -> polite out-of-scope refusal
Actual: (a) Asked clarifying question distinguishing Inbound vs Outbound OTD; (b) Politely refused weather query; 0 action tools called
Result: PASS

Row 5
Test: T5
Description: Risk + Recommendations
Expected: V_RISK_SIGNALS = 38 rows (all low priority, max score < 40, top = SUP-10U); CARR-01/03/07 flagged (>38.88%) & CARR-04 NOT flagged; RECOMMENDATIONS = 13 PROPOSED
Actual: 38 rows, all low (max = 36.83, top = SUP-10U); CARR-01/03/07 FLAGGED, CARR-04 NOT_FLAGGED (39.59%); exactly 13 PROPOSED rows (0 diff with candidate view)
Result: PASS

Row 6
Test: T6
Description: What-If Simulation Function
Expected: Pure table function with ::FLOAT args: Baseline (64.12, 0,0,0,0) -> 64.12 OTD, $0 net cost; Scenario (64.12, 15,20,7,10) -> 68.37 OTD, 26.58 risk, +$36,625 net cost; Extreme (64.12, 38,50,14,-20) -> OTD 5%le 100%, risk 5%ge 85
Actual: Baseline: 64.12% OTD, $0 net cost; Scenario: 68.37% OTD, 26.58% risk, +$36,625 net cost; Extreme: 88.62% OTD, 0.00% risk (clamping verified)
Result: PASS

Row 7
Test: T7
Description: Escalation Guardrail (Negative)
Expected: Calling SP_CREATE_ACTION_ITEM with invalid severity 'URGENT' returns error; APP.ACTIONS count remains 1
Actual: Returned "ERROR: Invalid severity: URGENT. Allowed values: LOW, MEDIUM, HIGH, CRITICAL."; APP.ACTIONS count = 1
Result: PASS

Row 8
Test: T8
Description: App + Safety State Verification
Expected: SHOW STREAMLITS lists SUPPLY_CHAIN_CONTROL_TOWER on HACK_WN; LIST @STREAMLIT_STAGE shows streamlit_app.py; 12M Outbound OTD = 66.12%; all 3 tasks = suspended; APP.ACTIONS = 1
Actual: Streamlit listed and active on HACK_WN; stage contains streamlit_app.py (41,920 bytes); 12M OTD = 66.12%; all 3 tasks suspended; APP.ACTIONS = 1
Result: PASS

Safety & Governance State Audit
• APP.ACTIONS Row Count:
  • Baseline (Before Suite): 1
  • Final (After Suite): 1
• Scheduled Tasks:
  • SC_ONTOLGOV_APP_TSH_ANOMALY_WATCHDOG: suspended
  • SC_ONTOLGOV_APP_TSH_DAILY_METRIC_CONSISTENCY_TESTS: suspended
  • SC_ONTOLGOV_GOVERNANCE_TSH_DRIFT-SENTINEL: suspended

All tests passed with zero data side effects or task reactivations.

[ ] Proceed to the app redeploy phase [Implementing Additive V2 Enhancements for Supply Chain Ontology]

>> confirm actions (shift+tab) | teams mode: off (ctrl+g) | plan mode: off (ctrl+p) | guardrails: off (shift+ctrl+s) Q zhsbcwu-tn70888
```

Screenshot: 36_CoCo_CLI_v2_tests_8of8_scorecard.png - T3-T8 results and safety audit: all PASS, APP.ACTIONS 1 -> 1, all tasks suspended.